

Lecture 06

Entrepreneurship in the Digital Age

Chapter lecture — 16 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XII · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — بارہویں جماعت

16 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XII

How to use this BIEK Computer Science lecture

Each PDF is a classroom lecture from the BIEK / Sindh Computer Science textbooks (New Sindh Curriculum, STBB Grade XI–XII). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before looking at the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر بی آئی کے / سندھ نصاب کی کتاب کے درس کے مطابق ہے۔ سبق پڑھیں، مثال حل کریں، پھر چیک یورسیلف کے سوالات کے جواب اپنے نوٹ بک میں لکھیں۔ کتاب کے بعد اضافی بورڈ پریکٹس ہے۔ ★ گولڈن ٹاپکس امتحان میں زیادہ پوچھے جاتے ہیں۔

Lectures in this pack

Chapter 6: Entrepreneurship in the Digital Age. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 6.1 Entrepreneur and Entrepreneurship

L03. Table 6.1 Local examples of digital entrepreneurship

L05. 6.4 Prototype

L07. 6.6 Prototype Development Cycle (Design, Build, Test, Iterate)

L09. 6.8 Minimum Viable Product (MVP)

L11. 6.10 Identifying the Riskiest Assumption

L13. 6.12 Case Study: MVP for a Canteen Pre-Order System

L15. 6.14 Beachhead Market

L02. 6.2 Entrepreneurship in the Digital Age

L04. 6.3 From Problem to Business Idea

L06. 6.5 Types of Prototypes (Low, Mid, High-Fidelity)

L08. 6.7 Class Activity: Create a Prototype

L10. 6.9 Difference between Prototype and MVP

L12. 6.11 Developing an MVP

L14. 6.13 Testing an MVP

L16. 6.15 & 6.16 Complete Student Project & Ethical Use

Entrepreneurship in the Digital Age

This chapter shows you how a real problem in your home, college or bazaar can become a business idea, and how you can test that idea cheaply with paper sketches, prototypes and a Minimum Viable Product (MVP) before you spend a lot of money. You will also learn how to choose your first users, the beachhead market, and how to collect feedback honestly and safely.

اس باب میں آپ سیکھیں گے کہ روزمرہ زندگی کے کسی حقیقی مسئلے کو کاروباری خیال میں کیسے بدلا جائے، اور پروٹوٹائپ (Prototype) اور ایم وی پی (MVP) کے ذریعے کم خرچ میں اس خیال کو کیسے آزمایا جائے۔

What you will learn

- Define **entrepreneur** and **entrepreneurship** and describe the habits of an entrepreneur.
- Explain how digital tools (WhatsApp, Google Forms, Facebook pages, simple websites) make it cheap and quick to start and test a business.
- Turn an everyday problem into a business idea using the steps Observe, Ask Users, Select One Problem, Suggest Solution, Test Early.
- Define a **prototype**, list its uses, and compare low-, mid- and high-fidelity prototypes.
- Draw and explain the prototype development cycle: Design, Build, Test, Iterate.
- Define a **Minimum Viable Product (MVP)** and state at least five differences between a prototype and an MVP.
- Identify the **riskiest assumption** of a business idea and design a small test for it.
- Prepare an MVP plan using the MVP planning checklist, and test the MVP with number-based and comment-based feedback.
- Choose a focused **beachhead market** for a new idea.
- Complete a small group project and use digital tools ethically, safely and fairly.

6.1 Entrepreneur and Entrepreneurship

LEARN IT

An **entrepreneur** is a person who **identifies a problem, thinks of a solution, and tries to change that solution into a product or service.**

An entrepreneur does not always begin with a large company or a big office. Many begin with a small idea, limited resources, and a strong wish to solve a problem. Think of a young person in Karachi who starts by delivering home-cooked lunch boxes to a few offices, or a student in Sukkur who repairs mobile phones for neighbours. They started small.

Entrepreneurs usually show three habits:

- **They observe people and listen to their needs.** They notice what annoys people again and again.
- **They take calculated risks.** A *calculated risk* is a risk you take after thinking about what you could lose and how to keep that loss small. It is not gambling.
- **They improve their idea step by step.** They do not expect the first attempt to be perfect.

What is entrepreneurship?

Entrepreneurship is the **process** of **planning, starting, testing, and improving** a business idea. It involves **creativity, problem-solving, teamwork, communication, and calculated risk.**

Notice the difference: the *entrepreneur* is the **person**; *entrepreneurship* is the **process** that person goes through.

REMEMBER IT

Simple meaning. An entrepreneur asks: "*What problem can I solve?*" Entrepreneurship is the process of testing and improving that solution until it becomes useful for people.

Remember the three-step journey:

1. Problem → 2. Solution → 3. Product

WORKED EXAMPLE

Worked example: spotting the three steps. Ayesha lives in Larkana. Her mother and other women in the street make beautiful hand-embroidered *rillis*, but they can only sell them to visitors.

Step 1, Problem: skilled women have products but very few buyers.

Step 2, Solution: show the rillis online so buyers from other cities can see them.

Step 3, Product/Service: a Facebook page with photos, prices and a "message to order" button.

Ayesha is the **entrepreneur**. Planning the page, posting photos, seeing which designs sell, and changing prices is **entrepreneurship**.

COMMON MISTAKE

Mistake: "Only rich people with big offices can be entrepreneurs."

Fix: The source clearly says many entrepreneurs begin with a small idea and limited resources. What matters is a real problem, a useful solution, and the willingness to improve it.

اردو میں سمجھیں

کاروباری شخص (Entrepreneur) وہ ہوتا ہے جو کسی مسئلے کو پہچانتا ہے، اس کا حل سوچتا ہے اور اس حل کو کسی پراڈکٹ یا سروس میں بدلنے کی کوشش کرتا ہے۔ کاروبار شروع کرنے، آزمانے اور بہتر بنانے کے پورے عمل کو انٹرپرائیورشپ (Entrepreneurship) کہتے ہیں۔ ایسا شخص لوگوں کی بات سنتا ہے، سوچ سمجھ کر خطرہ مول لیتا ہے اور اپنے خیال کو قدم بہ قدم بہتر بناتا ہے۔

CHECK YOURSELF

1. Is "entrepreneurship" a person or a process?
2. Name the three habits of an entrepreneur given in the source.
3. What is a calculated risk?

Answers: 1. A process (planning, starting, testing and improving a business idea). 2. Observing people and listening to their needs; taking calculated risks; improving the idea step by step. 3. A risk taken after thinking carefully about the possible loss and keeping that loss small.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An entrepreneur:

- a. Turns a problem into a product/service and takes the risk
- b. Only works in a bank job
- c. Only draws NAND
- d. Only writes OSI

MCQ 2. Entrepreneurship is the:

- a. Process of doing that (test and improve until useful)
- b. Logo
- c. First tweet
- d. Compiler

MCQ 3. A student can be entrepreneurial by:

- a. A small local solution (notes, tutoring, canteen queue)
- b. Waiting for a billion-dollar app
- c. Avoiding all users
- d. Skipping tests

MCQ 4. Risk in the book's sense includes:

- a. It might not work / no one pays
- b. NAND delay
- c. Median of 1
- d. OSI layer 2

MCQ 5. Not every shopkeeper in the bazaar is automatically:

- a. A digital entrepreneur (needs the digital-age piece)
- b. A failure
- c. A programmer
- d. A UX designer

MCQ 6. Ethics still apply:

- a. Don't scam, don't leak user data
- b. Scams are clever entrepreneurship
- c. Spy on classmates
- d. Copy a bank app's OTP

Extra short questions

1. Define entrepreneur and entrepreneurship.
2. A local non-app example.
3. Why "risk" is in the definition.
4. One quality of an entrepreneur from class discussion.

Extra long question

1. Is a class fellow who sells printed notes an entrepreneur? Use the book definitions, then say what would make it "digital age".

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Person vs process: problem → solution → offer to users, with risk.
2. A photocopy stall that also sells past papers at the gate — still entrepreneurship; digital comes next lecture.
3. You may waste time/money; users may not come.
4. Observes problems, tests, iterates — not just dreams.

Long

1. Yes if they identified a problem and run a small offer. Digital: WhatsApp catalogue, JazzCash, PDF store. Ethics: copyright of the textbook.

6.2 Entrepreneurship in the Digital Age ★ Golden

LEARN IT

Entrepreneurship in the digital age means using **digital tools** to **start, test, improve, and grow** a business idea.

These tools may include **mobile phones, social media, websites, online forms, digital payments, maps, videos, and simple apps**. In Pakistan you already know many of them: WhatsApp, Facebook, Instagram, TikTok, YouTube, Google Forms, Google Maps, and digital payments such as JazzCash, Easypaisa or Raast.

The power of digital tools

A digital entrepreneur **does not always need to create a complex mobile app**. Sometimes a **WhatsApp group**, a **Google Form**, a **Facebook page**, or a **simple website** is enough to test an idea.

In the past

- Large investment needed.
- Offices and shops had to be rented first.
- Long planning before anyone saw the product.
- A failed idea meant a big loss.

Today (digital age)

- A small idea can be tested **quickly**.
- Free tools: WhatsApp, Google Forms, Facebook.
- Feedback arrives in hours or days.
- If users do not like the idea, the team can **change it before spending too much money**.

WORKED EXAMPLE

Real-world relevance (from the source).

Example 1: A student selling **handmade cards through Instagram**. She posts photos of her Eid cards, people message her, and she delivers. No website, no app.

Example 2: A shopkeeper creating a **WhatsApp catalogue for Ajrak**. WhatsApp Business lets him list each Ajrak with a photo and price. Customers in Hyderabad or Karachi browse and order through chat.

Both are digital entrepreneurship **without writing a single line of code**.

Walk-through: how the Ajrak seller tests a new design cheaply.

1. He photographs 3 new Ajrak designs and adds them to the catalogue (cost: almost zero).
2. He sends the catalogue link to his customer broadcast list.
3. After one week he counts the orders per design.
4. He prints more of the design that sold most and stops the one nobody ordered.

He learned what customers want **before** printing a large stock.

REMEMBER IT

Remember "S-T-I-G": digital tools help you **Start, Test, Improve and Grow**. Think: "Start The Idea Growing."

COMMON MISTAKE

Mistake: "Digital entrepreneurship means building a mobile app."

Fix: An app is only one option. A WhatsApp group, Google Form, Facebook page or simple website is often enough, and much cheaper, to test whether people want your idea.

اردو میں سمجھیں

ڈیجیٹل دور میں انٹرپرائزورشپ کا مطلب بے موبائل فون، سوشل میڈیا، ویب سائٹ، آن لائن فارم اور ڈیجیٹل ادائیگی جیسے ٹولز کے ذریعے کاروبار شروع کرنا، آزمانا اور بڑھانا۔ ہر بار پیچیدہ ایپ بنانا ضروری نہیں؛ وائس ایپ گروپ یا گوگل فارم بھی کافی ہو سکتا ہے۔ اگر لوگوں کو خیال پسند نہ آئے تو زیادہ پیسہ خرچ ہونے سے پہلے اسے بدلا جا سکتا ہے۔

EXAM TIP

Common questions: "Define entrepreneurship in the digital age", "How have digital tools changed entrepreneurship?" and "Give examples of digital entrepreneurship without coding." Write the definition (start, test, improve, grow using digital tools), list 4–5 tools, give the past-versus-today contrast (big investment versus quick, cheap testing), and quote the Instagram cards and the WhatsApp Ajrak catalogue examples.

CHECK YOURSELF

1. List four digital tools that the source mentions.
2. Why is it easier to change a business idea today than in the past?
3. Give one example of digital entrepreneurship that needs no coding.

Answers: 1. Any four of: mobile phones, social media, websites, online forms, digital payments, maps, videos, simple apps. 2. Because a small idea can be tested quickly and cheaply, so if users do not like it the team can change it before spending too much money. 3. Selling handmade cards through Instagram, or a WhatsApp catalogue for Ajrak.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Digital-age entrepreneurship uses:

- a. Digital tools to start, test, improve, grow
- b. Only a physical shop with no phone
- c. Only NAND labs
- d. Only paper ledgers forever

MCQ 2. A WhatsApp business catalogue is:

- a. A digital channel
- b. OSI layer 1 hardware
- c. A primary key
- d. A K-map

MCQ 3. Low cost of testing online means:

- a. You can try an MVP cheaply
- b. You skip users
- c. You skip ethics
- d. You skip problems

MCQ 4. This is ★: define + two local examples + one risk.

- a. The recipe
- b. Only Silicon Valley names
- c. Only Python lists
- d. Only ER

MCQ 5. Payment via JazzCash is:

- a. Digital infrastructure an entrepreneur can use
- b. A sort algorithm
- c. A pie of NAND
- d. A Figma auto-layout

MCQ 6. Copying an app without a real user problem is:

- a. Not the book's method
- b. Guaranteed success
- c. A stack
- d. A median

Extra short questions

1. Define entrepreneurship in the digital age.
2. Two tools a student actually has.
3. A risk unique to digital.
4. Why local problems beat fake "Uber for X".

Extra long question

1. Take a photocopier stall. List 5 digital upgrades that fit Chapter 6 (not a full app). For each, the problem it tests. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Using digital tools to start/test/improve/grow a business.
2. Phone camera, WhatsApp, Google Form, Canva, GitHub pages — any two.
3. Cyber fraud, copycats, platform bans, data leak.
4. You can talk to users this week (canteen, hostels, tuition).

Long

1. WhatsApp queue, prepay JazzCash, Google Form for notes list, simple status stories, PDF sample. Each tests demand. Ethics: no pirated books.

Table 6.1 Local examples of digital entrepreneurship

LEARN IT

Digital entrepreneurship can begin from **ordinary problems**. The examples below are close to the context of Sindh and Pakistan, where many students, parents, and small service providers already use mobile phones.

Problem	Digital Idea	Possible First Users	Urdu Context
Parents miss college notices.	WhatsApp notice and reminder service.	Parents of first-year students.	والدین تک کالج کے نوٹس پہنچانے کے لیے واٹس ایپ سروس۔
Small embroidery shop has few customers.	Simple catalogue with order form on Facebook or WhatsApp.	Customers in the same city.	کڑھائی کی دکان کے لیے واٹس ایپ کیٹلاگ۔
Students forget test dates.	Google Calendar guide or simple reminder app.	Grade XII students.	ٹیسٹ کی تاریخیں یاد دلانے والی ایپ یا کیلنڈر۔
Canteen has long queues during break.	Pre-order form for snacks.	Students and teachers.	کینٹین کے لیے پری آرڈر فارم۔

Read each row from left to right: a **problem** → a **simple digital idea** → a **small group of first users**. You will see these same examples again later in the chapter (the canteen pre-order system becomes a full case study).

WORKED EXAMPLE

Worked example: adding your own row. Follow the same pattern for a new local problem.

Problem	Digital Idea	Possible First Users
Students in a Mirpurkhas college lose water bottles, calculators and ID cards and never find them.	Lost-and-found Google Form plus a WhatsApp group where found items are posted with a photo.	Students of one college.

Check: Is it a **real** problem? Yes, it happens every week. Is the idea **simple**? Yes, a form and a group. Are the first users **reachable**? Yes, they are in the same college.

COMMON MISTAKE

Important: A digital business idea should **not** be selected only because it sounds modern. It should solve a **real problem for real users**. If the problem is weak, the business idea will also be weak.

For example, "an AI app that chooses your shirt colour" sounds modern, but if nobody actually struggles with that, nobody will use it.

REMEMBER IT

Think "Problem first, tool second." Start with the pain, not with the technology.

اردو میں سمجھیں

جدول 6.1 بتاتی ہے کہ ڈیجیٹل کاروبار عام مسائل سے شروع ہو سکتا ہے، جیسے والدین کا کالج نوٹس نہ دیکھ پانا، کڑھائی کی دکان پر کم گاہک، ٹیسٹ کی تاریخ بھول جانا یا کینٹین پر لمبی قطار۔ کوئی خیال صرف اس لیے نہ چنیں کہ وہ جدید لگتا ہے؛ اسے حقیقی لوگوں کا حقیقی مسئلہ حل کرنا چاہیے۔

CHECK YOURSELF

1. In Table 6.1, what digital idea is suggested for a small embroidery shop?
2. Who are the possible first users of the canteen pre-order form?
3. Why should an idea not be chosen only because it sounds modern?

Answers: 1. A simple catalogue with an order form on Facebook or WhatsApp. 2. Students and teachers. 3. Because it must solve a real problem for real users; a weak problem gives a weak business idea.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Table 6.1 is there to show:

- a. Ordinary local problems can start digital businesses
- b. Only US startups count
- c. Only AI labs count
- d. Only OSI consultants count

MCQ 2. A good exam use of the table is:

- a. One example, the problem, the digital twist
- b. Copying 20 rows unread
- c. Ignoring Pakistan
- d. Only listing apps

MCQ 3. "Local" means:

- a. Your college/bazaar/city, not only Silicon Valley
- b. Only Karachi DHA
- c. Only English users
- d. Only GPUs

MCQ 4. If you invent an example it must still:

- a. Be realistic and ethical
- b. Promise billions
- c. Need NASA
- d. Need a compiler in C

MCQ 5. The table sits after 6.2 because:

- a. It illustrates digital-age entrepreneurship
- b. It replaces prototypes
- c. It is HCI only
- d. It is SQL only

MCQ 6. A rickshaw app idea should still start from:

- a. A real pain (wait, fare fights)
- b. A random blockchain
- c. A random NFT
- d. A random pie

Extra short questions

1. From memory, give two local digital entrepreneurship examples (book or equivalent).
2. For one example: problem → digital twist.
3. Why the board likes local examples.
4. One unethical "business".

Extra long question

1. Add three rows in Table 6.1 style for your college: problem, idea, digital tool, first test.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Food delivery, online tuition, used-books WhatsApp, ride apps, handicraft Instagram — mapped to a problem.
2. Long canteen line → pre-order form.
3. Shows you understood 6.2, not just American brand names.
4. Selling leaked papers; fake job ads.

Long

1. Queue, lost keys, hostel laundry. Tools: form, group, JazzCash. First test: 10 users this week. Ethics column.

6.3 From Problem to Business Idea

LEARN IT

A good business idea usually starts from **observing daily life**. Look around your home, college, market, village, or city and ask these simple questions:

1. **What problem do people face?**
2. **Who faces this problem?**
3. **How often does it happen?**
4. **What do people currently do to solve it?**
5. **Can technology help in a simple way?**

A short way to remember them: **What? Who? How often? Current fix? Tech help?**

WORKED EXAMPLE

Example scenario (from the source): study material before exams.

Students cannot find study material before exams. The quick answer is "students are careless", but that is not the real issue.

Apply the five questions:

1. **What problem?** Students cannot find notes before exams.
2. **Who?** Students of a class or college.
3. **How often?** Before every test and exam.
4. **Current fix?** Notes are shared in many places: WhatsApp, email, photocopies, and class groups. Students search all of them.
5. **Tech help?** Yes: a simple **subject-wise notes organizer** where all files are placed in one location (for example, one shared Google Drive folder per subject).

The real problem is **scattered notes**, and the idea solves exactly that.

Scope of the problem

Avoid choosing **very large problems** at the start, such as "improving the whole education system". A better problem is **smaller and clear**, such as helping Grade XII students track **practical file submission dates**. Smaller problems are **easier to test and improve**.

Table 6.2: From problem to idea

The journey from observing a problem to creating an idea follows a structured path.

Step	What Students Do	Example
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Observe	Notice repeated problems around home, college, or market.	Students forget practical file dates.
Ask Users	Talk to people who face the problem.	Ask five classmates how they remember deadlines.
Select One Problem	Choose one clear problem only.	Practical file reminder.
Suggest Solution	Think of a simple digital or paper-based solution.	Reminder form or app screen.
Test Early	Show the idea to users before final development.	Ask classmates if they will use it.

After this, the idea moves along the full path that the rest of this chapter follows:



REMEMBER IT

Mnemonic for Table 6.2: "Only Ask Simple Solutions Today": Observe, Ask users, Select one problem, Suggest solution, Test early.

COMMON MISTAKE

Mistake 1: Choosing a huge problem ("fix traffic in Karachi"). **Fix:** Narrow it ("help students of one college know when their van will arrive").

Mistake 2: Skipping "Ask Users" and guessing what people need. **Fix:** Talk to at least five real people who face the problem.

اردو میں سمجھیں

اچھا کاروباری خیال روزمرہ زندگی کے مشاہدے سے شروع ہوتا ہے۔ پوچھیں: مسئلہ کیا ہے، کس کو ہے، کتنی بار ہوتا ہے، لوگ ابھی کیا کرتے ہیں، اور کیا ٹیکنالوجی آسان طریقے سے مدد کر سکتی ہے؟ شروع میں چھوٹا اور واضح مسئلہ چنیں، کیونکہ اسے آزمانا اور بہتر بنانا آسان ہوتا ہے۔

CHECK YOURSELF

1. Write the five steps of Table 6.2 in order.
2. Why is "improving the whole education system" a poor first problem?
3. In the study-material example, what is the real problem?

Answers: 1. Observe, Ask Users, Select One Problem, Suggest Solution, Test Early. 2. It is too large and unclear; small, clear problems are easier to test and improve. 3. Notes are scattered in many places (WhatsApp, email, photocopies, class groups), not that students are careless.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.

2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.

6.4 Prototype ★ Golden

LEARN IT

A **prototype** is an **early model** of a product, service, website, or app. It shows **how the idea may look or work**. It is **NOT the final product**. It is made to **test the idea and collect feedback**.

Think of the clay model an architect makes before building a house, or the sample *kameez* a tailor stitches before making 50 for a shop. The model lets people say "make this bigger" or "move that here" before real money is spent.

Why is a prototype useful?

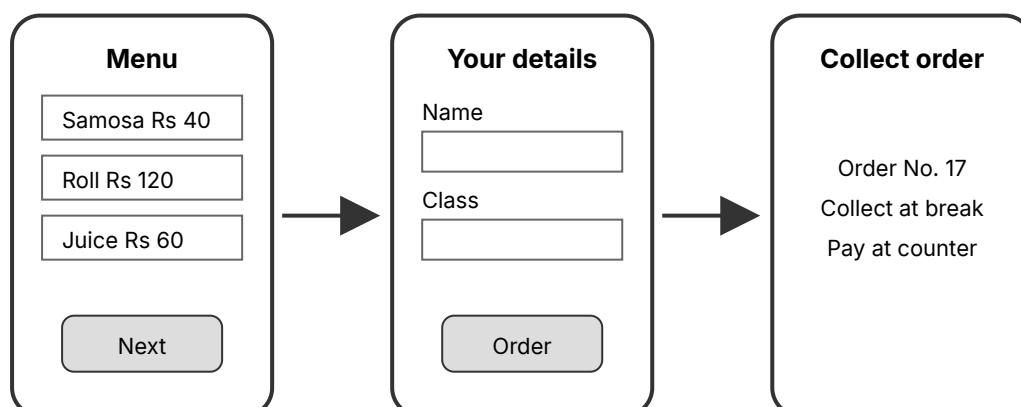
A prototype helps students and entrepreneurs **explain their idea clearly**, because users can **see something** instead of only listening to words.

- **It makes the idea visible.**
- **It helps the team find mistakes early.** Users can point out if a button is hard to find, if a screen has too much text, or if a feature is missing.
- **It saves time and money** before final coding begins. Fixing a pencil sketch takes a minute; fixing finished code can take days.

What form can a prototype take?

A prototype may be a **paper sketch**, a **cardboard model**, a **clickable PowerPoint**, a **Canva design**, a **Figma wireframe**, a **simple website layout**, or a **sample service process chart**. A *wireframe* is a simple outline drawing of a screen that shows where things go (boxes, buttons, text) without colours or pictures.

For example, for a **canteen pre-order system**, students can just **draw the screens on paper**.



A paper prototype of a canteen pre-order system: three hand-drawn screens joined by arrows (select item → enter name → collect order).

WORKED EXAMPLE

Worked example: how a paper prototype finds a mistake early. Bilal's team in a Nawabshah college draws the three screens above on A4 paper.

1. They hand screen 1 to a classmate and say, "Order a samosa."
2. The classmate looks for a quantity box and cannot find one. She asks, "What if I want two?"
3. The team notes: **feature missing: quantity.**
4. They add a small "Qty: []" box with a pencil in 30 seconds.

If they had already coded an app, adding quantity would have meant changing the database, the screen and the order list. The prototype **saved time and money.**

REMEMBER IT

Prototype = "practice model". It is like a rough draft of an essay: you show it, get corrections, and rewrite it before the final copy.

COMMON MISTAKE

Mistake: Thinking a prototype must work or must look beautiful.

Fix: A prototype only has to make the idea **visible** so users can react. A pencil sketch is a valid prototype.

اردو میں سمجھیں

پروٹوٹائپ (Prototype) کسی پراڈکٹ، سروس، ویب سائٹ یا ایپ کا ابتدائی نمونہ ہوتا ہے جو دکھاتا ہے کہ خیال کیسا نظر آئے گا یا کیسے کام کرے گا۔ یہ حتمی پراڈکٹ نہیں ہوتا بلکہ خیال کو آزمانے اور رائے (Feedback) لینے کے لیے بنایا جاتا ہے۔ یہ کاغذ کا خاکہ، گتے کا ماڈل، پاورپوائنٹ یا فگما (Figma) ڈیزائن بھی ہو سکتا ہے، اور اس سے غلطیاں جلد پکڑی جاتی ہیں اور وقت اور پیسہ بچتا ہے۔

EXAM TIP

Expect "Define prototype and write its benefits" (3–5 marks). Write: (1) definition with the words *early model, not the final product, test the idea and collect feedback*; (2) three benefits: makes the idea visible, finds mistakes early, saves time and money; (3) forms it can take (paper sketch, cardboard model, clickable PowerPoint, Canva, Figma wireframe); (4) the canteen paper-screens example.

CHECK YOURSELF

1. Is a prototype the final product?
2. Name four forms a prototype can take.
3. Give two kinds of problems users can point out by looking at a prototype.

Answers: 1. No. It is an early model made to test the idea and collect feedback. 2. Any four: paper sketch, cardboard model, clickable PowerPoint, Canva design, Figma wireframe, simple website layout, sample service process chart. 3. Any two: a button is hard to find, a screen has too much text, a feature is missing.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A prototype is:

- a. An early model of how it may look/work — not the final product
- b. The finished mass-produced item
- c. A database backup
- d. A pie of revenue

MCQ 2. You build it to:

- a. Learn, not to impress investors first
- b. Win a logo contest only
- c. Replace users
- d. Skip tests

MCQ 3. This is ★: define, purpose, what it is NOT.

- a. The 5-mark
- b. Only draw Figma pretty
- c. Only Python
- d. Only OSI

MCQ 4. A paper phone screen is still a:

- a. Prototype (low fidelity)
- b. MVP already charged
- c. Compiler
- d. Firewall

MCQ 5. Prototype before code when:

- a. The question is "do people understand the idea?"
- b. You already have 1 million users
- c. The law froze the spec like waterfall NASA
- d. You are only drawing NAND

MCQ 6. Throwing a prototype away can be:

- a. Success (you learned cheaply)
- b. Shame
- c. A SQL error
- d. A median

Extra short questions

1. Define prototype from the book.
2. Three purposes.
3. What it is not.
4. A non-app prototype.

Extra long question

1. Design a paper prototype of a "lost-and-found" college app: 3 screens, 3 things you would watch in a test. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Early model of a product/service/app that shows how it may look or work. Not the final thing.
2. Communicate idea, test with users, find problems early.
3. Not the finished product; not necessarily coded; not an MVP (next topics).
4. A cardboard token for a queue system.

Long

1. Home, report item, search. Watch: do they find report in 30s, do they understand photos, do they trust privacy. Don't code yet.

6.5 Types of Prototypes (Low, Mid, High-Fidelity)

LEARN IT

Prototypes are commonly divided into **three types** according to their level of detail, called **fidelity**. **Fidelity means how close the prototype is to the final product.** Low fidelity = far from final; high fidelity = very close to final.

6.5.1 Low-Fidelity Prototype

A **quick and rough** model made at the **very beginning**. It is **drawn on paper**. It does **not focus on colours or fonts**. It shows the **main idea and flow** (which screen comes after which). It is **very suitable for students** because it is **cheap, fast, and easy to change**.

6.5.2 Mid-Fidelity Prototype

Clearer than a paper sketch. It is created in **PowerPoint, Canva, or Figma**. It shows screens and **navigation** in a more organised way, but **without final graphics**. It is useful to **test screen-to-screen navigation**, for example by linking slides so that clicking "Next" opens the next screen.

6.5.3 High-Fidelity Prototype

Looks **very close to the final product**. It includes **realistic colours, buttons, sample data, and working features**. It **takes more time** and should **only be created after the basic idea is tested**.

Comparison of prototype types

For most school and college projects, you should **begin with a low-fidelity prototype**. A common mistake is to start building a full app before knowing whether users need it. A rough prototype protects you from this mistake.

Table 6.3: Types of prototypes

Type	Meaning	Best Use	Example
Low-Fidelity	A quick and rough model of the idea.	Very early testing.	Hand-drawn app screens for homework reminder.
Mid-Fidelity	A clearer model that shows layout and flow.	Testing navigation.	Clickable screens in PowerPoint or Figma.
High-Fidelity	A detailed model close to the final product.	Testing realistic experience.	Working demo of a small order form website.

Low-Fi (Paper) → Mid-Fi (PowerPoint) → High-Fi (Figma/HTML)

Extra comparison to help you revise:

Feature	Low-Fidelity	Mid-Fidelity	High-Fidelity
Tool	Paper and pencil	PowerPoint, Canva, Figma	Figma (detailed), HTML
Colours and fonts	No	Not final	Realistic
Clickable?	No (you move papers by hand)	Yes, basic links	Yes, with working features
Cost and time	Lowest, fastest	Medium	Highest, slowest
Easy to change?	Very easy	Fairly easy	Harder
When	Very beginning	After the flow is agreed	Only after the basic idea is tested

WORKED EXAMPLE

Worked example: one idea, three fidelities. Idea: a Homework Reminder for a college in Jacobabad.

1. **Low-fi:** Sana draws three screens on paper in 15 minutes: "Today", "Add homework", "Reminder". Classmates point at the drawings to "use" them.
2. **Mid-fi:** After feedback, she makes three PowerPoint slides with grey boxes and links them with hyperlinks, so tapping "Add" opens the Add slide. Now she tests whether people can move between screens easily.
3. **High-fi:** Once users clearly want it, she builds a small HTML page with real colours, a working form and sample homework data.

Each step costs more, so she only moves up when the previous step proves the idea is worth it.

REMEMBER IT

Fidelity = "faithfulness" to the final product. Remember **Paper** → **PowerPoint** → **Polished** (low → mid → high).

COMMON MISTAKE

Mistake: Starting with a high-fidelity prototype or a full app because it "looks professional".

Fix: Begin with low fidelity. If users do not need the idea, you have lost only paper and an hour, not weeks of work.

فِڈیلیٹی (Fidelity) کا مطلب ہے کہ پروٹوٹائپ حتمی پراڈکٹ سے کتنا ملتا جلتا ہے۔ لو فِڈیلیٹی کاغذ پر بنایا گیا سستا اور جلدی بننے والا خاکہ ہے؛ مڈ فِڈیلیٹی پاورپوائنٹ، کینوا یا فگما میں بنتا ہے اور اسکرینوں کے درمیان آنا جانا (Navigation) دکھاتا ہے؛ ہائی فِڈیلیٹی بالکل اصل پراڈکٹ جیسا لگتا ہے اور اسے بنیادی خیال آزمانے کے بعد ہی بنانا چاہیے۔

CHECK YOURSELF

1. What does "fidelity" mean?
2. Which type of prototype is best for testing navigation?
3. Why should students begin with low fidelity?

Answers: 1. How close the prototype is to the final product. 2. Mid-fidelity (e.g. clickable screens in PowerPoint or Figma). 3. It is cheap, fast and easy to change, and it protects them from building a full app before knowing whether users need it.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.

2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

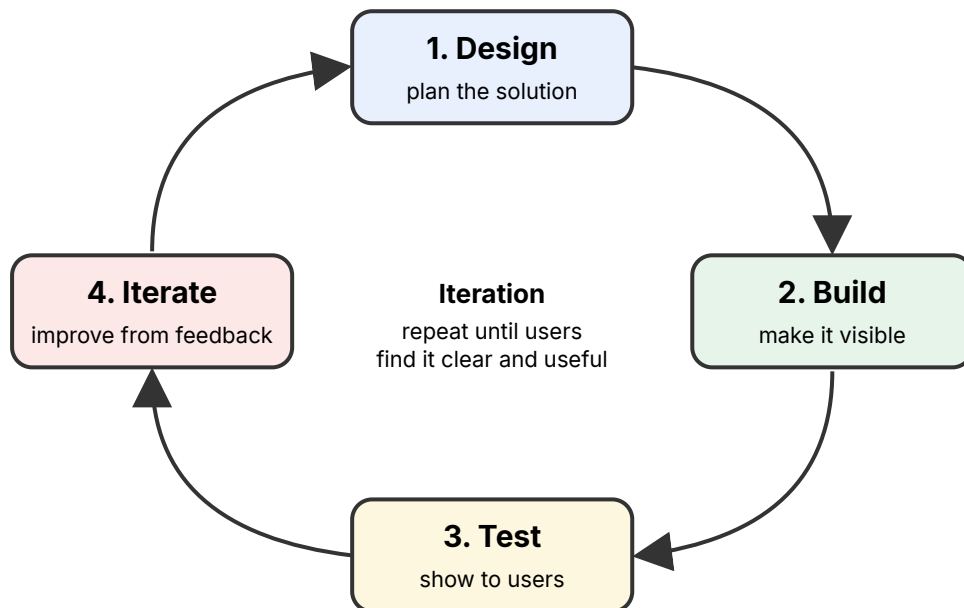
1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.

6.6 Prototype Development Cycle (Design, Build, Test, Iterate)

★ Golden

LEARN IT

Prototype development is **not a one-time activity**. A prototype is **created, tested, improved, and tested again**. This repeating process is called **iteration**. The main stages are **Design, Build, Test, and Iterate**.



The prototype development cycle: Design → Build → Test → Iterate, then back to Design.

Stage	What happens
1. Design	Understand the problem and plan the solution. Decide what the prototype should show (e.g. screen flow, features).
2. Build	Create the first version (physical or digital). The aim is not beauty; the aim is to make the idea visible.
3. Test	Show the prototype to users. Observe what they understand and what they find difficult.
4. Iterate	Improve the prototype based on user feedback. Remove unnecessary parts and add missing features.

WORKED EXAMPLE

Worked example: two rounds of the cycle. A team in Thatta wants a WhatsApp catalogue for a local Ajrak shop.

Round 1

- **Design:** The problem is that customers do not know which designs and prices are available. Plan: one page per design with photo, price and an "Order" message.
- **Build:** They make a paper booklet with 6 hand-drawn "catalogue pages".
- **Test:** They show it to 5 customers. Three ask, "Is this single or double-sided Ajrak? What size?"
- **Iterate:** They add size and type to every page.

Round 2

- **Design:** Add the size and type fields, and a delivery note.
- **Build:** They make the catalogue in WhatsApp Business with real photos.
- **Test:** 5 new customers browse it. Now nobody asks about size; one says the delivery charge is unclear.
- **Iterate:** They add "Delivery in Thatta: Rs 100" to each item.

Each round produces a clearer version, because each change comes from real feedback.

REMEMBER IT

Mnemonic: "Do Big Things Iteratively" = Design, Build, Test, Iterate. And remember the arrow at the end goes **back to Design**: it is a **cycle**, not a line.

COMMON MISTAKE

Mistake 1: Drawing the stages as a straight line that ends at "Iterate". **Fix:** Always draw a loop with an arrow from Iterate back to Design.

Mistake 2: Making the Build stage beautiful. **Fix:** The source says "The aim is not beauty; the aim is to make the idea visible."

اردو میں سمجھیں

پروٹوٹائپ ایک ہی بار میں نہیں بنتا؛ اسے بنایا، آزمایا، بہتر کیا اور دوبارہ آزمایا جاتا ہے۔ اس بار بار دہرانے کے عمل کو **ٹریشن (Iteration)** کہتے ہیں۔ چار مراحل ہیں: ڈیزائن (مسئلہ سمجھ کر حل کی منصوبہ بندی)، بلڈ (پہلا نمونہ بنانا)، ٹیسٹ (صارفین کو دکھانا) اور **ٹریٹ** (رائے کی بنیاد پر بہتری)۔

EXAM TIP

A favourite long question: "Explain the prototype development cycle with a diagram." Draw the four boxes in a **circle** with arrows, write one or two lines for each stage using the source's words, define **iteration**, and finish with a short example (e.g. the ClassBuddy activity in 6.7). Marks are often lost for drawing a straight line instead of a cycle.

CHECK YOURSELF

1. What is iteration?
2. At which stage do you observe what users find difficult?
3. What is the aim of the Build stage?

Answers: 1. The repeating process of creating, testing, improving and testing a prototype again. 2. The Test stage. 3. To make the idea visible (not to make it beautiful).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The cycle is:

- a. Design → build → test → iterate (repeat)
- b. Build once, never look
- c. Test on no one
- d. Advertise first

MCQ 2. Iterate means:

- a. Change the prototype from what you learned
- b. Print more logos
- c. Hire 50 people
- d. Write OSI parsers

MCQ 3. This is ★ because:

- a. Entrepreneurship is not one-shot in the book
- b. Waterfall never iterates even in software
- c. Users are optional
- d. Ethics are optional

MCQ 4. If tests fail, you:

- a. Return to design
- b. Ship anyway always
- c. Delete the users
- d. Change the syllabus

MCQ 5. A cycle can be:

- a. Hours (paper) not months
- b. Only yearly
- c. Only after IPO
- d. Only in Silicon Valley

MCQ 6. Skipping test in the cycle turns it into:

- a. Just building in the dark
- b. HCI gold
- c. SQL gold
- d. Big O gold

Extra short questions

1. Name the four stages.
2. What do you carry from test to design?
3. How many cycles is "enough" for class?
4. Link to Agile from XI.

Extra long question

1. Document two cycles of a homework-reminder prototype. Each cycle: what you built, who tested, what changed. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Design, build, test, iterate.
2. Notes: where people stuck, what they asked for, what they ignored.
3. At least two documented rounds.
4. Same spirit: small loops with feedback. Don't over-claim they are identical.

Long

1. Cycle1 paper: people wanted SMS not email. Cycle2 Figma with SMS mock. Metric: 4/5 could complete. Table.

6.7 Class Activity: Create a Prototype

LEARN IT

Business idea scenario. Students often forget **homework deadlines, test dates, and required class items**. A group of students wants to design a simple "**Homework Helper**" or "**ClassBuddy**" reminder system.

The team will **not** immediately build a complete mobile app. They will first **understand the problem, draw a low-fidelity prototype, test it with classmates, and improve it**. This activity is the development cycle of 6.6 in action.

1. Understand → 2. Draw low-fi → 3. Test → 4. Analyse feedback → 5. Improve

Step 1: Understand the problem

The team **interviews ten students**. They find that **7 out of 10** students (that is 70%) forget at least one task in a week. Students want **reminders before leaving home**, and some prefer a **very simple design**.

Table 6.4: Understanding user needs

Finding	Meaning	Design Decision
7 out of 10 forget tasks.	The problem is common.	App should show a daily task list.
Reminders needed before leaving home.	Reminder time is important.	Add morning notification option.
Students want simple apps.	Design must be simple.	Use large buttons and clear labels.

Step 2: Draw a low-fidelity prototype

The team draws **three screens on paper**: (1) **Today's classes**, (2) **Add homework**, (3) **Show reminder**. **Arrows** show how a user moves from one screen to another.

Step 3: Test the prototype

The team gives the paper screens to **five classmates**. The **observer watches silently** to see if users can understand the design by themselves. Users are given one task: "**add one homework task**".

Step 4: Analyse feedback

The team looks for **repeated problems** in the feedback to identify what actually needs fixing.

Table 6.5: Changing feedback into improvement

User Feedback	Problem Identified	Improvement
"I cannot quickly find Add Homework."	Button is not visible.	Make Add Homework button larger and centre it.
"I want reminders by subject."	Reminder is too general.	Add subject name and due date fields.
"I want to see all tasks for the week."	Only daily view is not enough.	Add a 'Weekly View' button.

Step 5: Improve the prototype

The team **redraws the home screen** with a **larger Add Homework button**, a **subject list**, and a **weekly view**. This second version is clearer because it is based on real user feedback.

A prototype has not failed when users give negative feedback; it only fails if the team ignores useful feedback.

WORKED EXAMPLE

Worked example: link the activity to the cycle.

Cycle stage	ClassBuddy activity step
Design	Step 1: interview 10 students, make design decisions (Table 6.4).
Build	Step 2: draw three paper screens with arrows.
Test	Step 3: five classmates try "add one homework task" while the observer watches silently.
Iterate	Steps 4 and 5: analyse feedback (Table 6.5) and redraw the home screen.

Why watch silently? If the observer explains ("tap here"), the team never learns that the button is hard to find. Silence shows the real difficulty.

REMEMBER IT

Feedback → Problem → Improvement. Every comment should be turned into a problem statement and then one clear change, exactly like the three columns of Table 6.5.

COMMON MISTAKE

Mistake: Feeling that negative feedback means the prototype "failed", or changing the design for every single comment.

Fix: Negative feedback is useful information. Look for **repeated** problems and fix those first. The prototype only fails if you ignore useful feedback.

اردو میں سمجھیں

اس سرگرمی میں طلبہ "کلاس بڈی" (ClassBuddy) نامی یاد دہانی نظام بناتے ہیں۔ پہلے دس طلبہ سے بات کر کے مسئلہ سمجھتے ہیں، پھر کاغذ پر تین اسکرینیں بناتے ہیں، پانچ ساتھیوں سے خاموشی سے آزمواتے ہیں، رائے کا تجزیہ کرتے ہیں اور ڈیزائن بہتر کرتے ہیں۔ منفی رائے سے پروٹوٹائپ ناکام نہیں ہوتا؛ ناکامی تب ہے جب مفید رائے کو نظرانداز کیا جائے۔

CHECK YOURSELF

1. In Step 1, how many students forgot at least one task in a week, and what percentage is that?
2. Name the three paper screens drawn in Step 2.
3. What improvement was made for the feedback "I want to see all tasks for the week"?

Answers: 1. 7 out of 10, which is 70%. 2. Today's classes, Add homework, Show reminder. 3. A 'Weekly View' button was added.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The class activity exists to:

- a. Make you build and test a prototype, not only define it
- b. Replace the exam
- c. Teach NAND
- d. Teach OSI

MCQ 2. Forgotten deadlines is a:

- a. Problem you can prototype
- b. GPU problem
- c. Primary key problem
- d. K-map problem

MCQ 3. Five classmates as testers are:

- a. Enough for a class test if you watch silently
- b. A statistically perfect census
- c. Useless
- d. A DoS attack

MCQ 4. The observer should:

- a. Watch, not rescue immediately
- b. Click for them
- c. Shout answers
- d. Grade their fashion

MCQ 5. You record:

- a. Where they hesitate and what they say
- b. Only a selfie
- c. Only the logo poll
- d. Only Big O

MCQ 6. After the activity you still:

- a. Iterate
- b. Ship to Play Store same hour
- c. Skip ethics
- d. Skip the write-up

Extra short questions

1. Summarise the book's activity scenario.
2. Three paper screens you would draw.
3. How to test without coding.
4. One ethical rule in class tests.

Extra long question

1. Write a lab report of the activity: idea, 3 screens, 5-user results, 3 changes. Use the cycle words.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Students forget homework/tests/items; team prototypes a reminder.
2. Today's tasks; add task; alarm confirm.
3. Ask a classmate to "use" paper as if it were a phone; you swap sheets.
4. Don't mock testers; don't collect their personal timetable into a public group.

Long

1. Results: 2/5 missed Add. Changes: bigger button, Urdu, confirm. Iterate. Marks-style headings.

6.8 Minimum Viable Product (MVP) ★ Golden

LEARN IT

A **Minimum Viable Product (MVP)** is the **simplest working version** of a product. It includes **only the most important features** needed to **test the idea with real users**.

- **Minimum:** it has only the **essential** features. No extras.
- **Viable:** it is **usable enough** to actually solve the main problem. ("Viable" means able to work successfully.)

An MVP is **not the final product**. It is also **not only a drawing**. It is a **small working version** used to test whether users actually need the solution.

WORKED EXAMPLE

Example from the source: canteen app.

Complete canteen app (final product)	Canteen MVP
Login, online payment, delivery tracking, discount offers...	A simple Google Form where students select snacks before break.

Step-by-step: how to cut a big idea down to an MVP.

1. List all features you dream of: login, payment, delivery tracking, discounts.
2. Ask of each one: "Without this, can a student still pre-order a snack?"
3. Login? Not needed; the student can type name and class. Payment? Not needed; pay at the counter. Delivery tracking? Not needed; students collect. Discounts? Not needed.
4. What is left is the core: **choose snack + name + class**. A Google Form does this for free.

That form is **minimum** (no extras) and **viable** (students really can pre-order with it).

REMEMBER IT

Simple meaning. An MVP answers: "**Will people actually use this solution?**" It tests **real use and behaviour**, not only the visual design.

Analogy: before opening a full restaurant, a cook in Hyderabad sells biryani from home to office workers for two weeks. If people keep ordering, the idea is worth growing. That home kitchen is the MVP.

COMMON MISTAKE

Mistake 1: Calling a paper sketch an MVP. **Fix:** A sketch is a prototype. An MVP must **work** and be used by **real users**.

Mistake 2: Thinking MVP means a low-quality or broken product. **Fix:** It is small but **viable**; it must genuinely solve the main problem.

اردو میں سمجھیں

ایم وی پی (Minimum Viable Product) کسی پراڈکٹ کا سب سے سادہ مگر کام کرنے والا ورژن ہے جس میں صرف ضروری خصوصیات ہوتی ہیں۔ "منیمم" کا مطلب ہے صرف ضروری چیزیں، اور "وائیل" کا مطلب ہے کہ وہ اصل مسئلہ حل کرنے کے قابل ہو۔ یہ حتمی پراڈکٹ نہیں بلکہ یہ جانچنے کا ذریعہ ہے کہ کیا لوگ واقعی اسے استعمال کریں گے، جیسے کینٹین کے لیے ایک سادہ گوگل فارم۔

EXAM TIP

Typical questions: "What is an MVP? Explain the words Minimum and Viable" and "Give an example of an MVP." Write the full form, the definition (simplest working version, only most important features, test with real users), explain both words, state that it is neither the final product nor only a drawing, and give the canteen Google Form example. Mention the key question: "Will people actually use this solution?"

CHECK YOURSELF

1. Write the full form of MVP.
2. What does "Viable" mean in MVP?
3. What question does an MVP answer?

Answers: 1. Minimum Viable Product. 2. Usable enough to actually solve the main problem. 3. "Will people actually use this solution?"

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An MVP is:

- a. The simplest working version that can test the riskiest assumption
- b. A paper sketch only
- c. The full app with every feature
- d. A logo

MCQ 2. Working means:

- a. Real users can get the core value (even if ugly/manual)
- b. It compiles Hello World
- c. It has 40 screens
- d. It has AI always

MCQ 3. This is ★: contrast with prototype in the next lecture, but here emphasise:

- a. Learning from a real offer
- b. Colours
- c. NAND
- d. OSI

MCQ 4. A WhatsApp group that actually takes canteen orders is:

- a. Closer to MVP than a Figma only
- b. Not digital
- c. A stack
- d. A pie

MCQ 5. Building extra features first is:

- a. The usual MVP failure
- b. Required
- c. Agile gold always
- d. HCI gold always

MCQ 6. The goal is to learn:

- a. Whether people use/pay, cheaply
- b. Whether the CEO likes purple
- c. Whether Big O is n^2
- d. Whether ER is 3NF

Extra short questions

1. Define MVP from the book.
2. What "viable" means.
3. What "minimum" means.
4. One example.

Extra long question

1. Design an MVP for "notes exchange" that you could run next week without an app store. List in/out features and the learning question. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Minimum Viable Product: simplest working version with only essential features, used to learn from real users.
2. Someone can complete the core job (order food, book notes).
3. Strip extras (reviews, AI, dark mode) until the core is tested.
4. Google Form + JazzCash + pickup window, before an app.

Long

1. In: list of notes, request, pickup. Out: chat, ratings, payments maybe later. Question: will 20 students actually exchange? How you measure.

6.9 Difference between Prototype and MVP

LEARN IT

Prototype and MVP are **related, but not the same**. A **prototype** is mainly used to **show and test the design** of an idea. An **MVP** is used to **test real use and market demand**. *Market demand* means whether enough people want the product and will use (or pay for) it.

Usually a prototype comes first and an MVP comes after it: Problem → Idea → Prototype → Feedback → MVP.

Table 6.6: Prototype and MVP comparison

Point	Prototype	MVP
Main purpose	To show how the idea may look or work.	To test whether users will actually use the solution.
Working status	May not work; can be a sketch or model.	Must work at least in a basic way.
Users	Small group gives feedback on design.	Early real users test the product in daily life.
Cost	Very low cost.	Low cost but may need some basic development.
Example	Paper screens for a canteen order system.	Google Form used for real canteen orders for three days.

REMEMBER IT

Key takeaway

Prototype = "Does it **look right** and make sense?"

MVP = "Will people **actually use it** and find value in it?"

Memory hook: **Prototype** → **P**icture of the idea; **MVP** → **M**arket test of the idea.

WORKED EXAMPLE

Worked example: classify each item. Decide whether each is a prototype or an MVP and give the reason.

1. Hand-drawn screens of a tuition-finder app shown to 6 friends. → **Prototype**: it does not work; it tests design.
2. A WhatsApp broadcast group that really sends van arrival times to parents of one route for a week. → **MVP**: it works and real users use it daily.
3. Clickable PowerPoint screens of a notes organizer. → **Prototype** (mid-fidelity): links work, but nobody uses it for real notes.
4. A shared Google Drive folder per subject that a class actually uses before the monthly test. → **MVP**: real use in daily life.

Rule used: **Does it work, and is it used by real users in real life?** If yes → MVP. If it only shows the design → prototype.

COMMON MISTAKE

Mistake: Thinking "high-fidelity prototype" and "MVP" are the same because both look good.

Fix: The difference is **purpose and real use**, not appearance. A beautiful Figma design that nobody uses for real orders is still a prototype; a plain Google Form taking real orders is an MVP.

اردو میں سمجھیں

پروٹوٹائپ اور ایم وی پی آپس میں جڑے ہیں مگر ایک جیسے نہیں۔ پروٹوٹائپ خیال کا ڈیزائن دکھاتا اور آزماتا ہے اور ضروری نہیں کہ کام کرے، جبکہ ایم وی پی کم از کم بنیادی طور پر کام کرتا ہے اور اصل صارفین روزمرہ زندگی میں اسے استعمال کرتے ہیں۔ پروٹوٹائپ پوچھتا ہے "کیا یہ ٹھیک لگتا ہے؟" اور ایم وی پی پوچھتا ہے "کیا لوگ واقعی اسے استعمال کریں گے؟"

CHECK YOURSELF

1. Which one must work at least in a basic way?
2. Give the source's example of a prototype and of an MVP for the canteen.
3. Which question does a prototype answer?

Answers: 1. The MVP. 2. Prototype: paper screens for a canteen order system. MVP: a Google Form used for real canteen orders for three days. 3. "Does it look right and make sense?"

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A prototype mainly:

- a. Shows/tests the design
- b. Takes real payments at scale always
- c. Replaces the company
- d. Is the beachhead

MCQ 2. An MVP mainly:

- a. Delivers the core working offer to real users to learn
- b. Is always paper
- c. Is always high-fi Figma only
- d. Has every feature

MCQ 3. You can have a prototype that is:

- a. Not working software
- b. Already the full product
- c. A DBMS
- d. An OSI stack

MCQ 4. You should not call a clickable mock the:

- a. MVP (if nobody can actually complete the job)
- b. Prototype
- c. Wireframe
- d. Test

MCQ 5. Order often is:

- a. Need → prototype → MVP → later product
- b. MVP → ignore users → prototype
- c. IPO → problem
- d. GPU → problem

MCQ 6. Exam 5-mark: table of:

- a. Purpose, working?, users, fidelity, example
- b. Only definitions swapped
- c. Only careers
- d. Only Python

Extra short questions

1. Four differences prototype vs MVP.
2. Can an MVP look ugly?
3. Can a prototype be digital?
4. One sentence you should memorise.

Extra long question

1. Table + one paragraph: for the canteen idea, what is the prototype and what is the MVP. Do not mix them.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Purpose design vs learning from a real offer; may not work vs must work; testers vs customers; cheap fake vs minimum real.
2. Yes — viability $\neq$ beauty.
3. Yes (Figma) — still not MVP if it doesn't deliver the job.
4. Prototype answers "do they understand it?"; MVP answers "do they use it?".

Long

1. Prototype: paper/Figma order flow. MVP: form that cooks actually receive and food is collected. Learning: % of orders picked up.

6.10 Identifying the Riskiest Assumption

LEARN IT

Every business idea is based on **assumptions**. An **assumption** is something we **believe to be true, but have not tested yet**.

The **riskiest assumption** is the belief that can make the **entire business fail** if it turns out to be wrong.

Entrepreneurs should **identify and test the riskiest assumption early**, because it **saves time and money**. If it is wrong, you find out before building everything else.

Table 6.7: Testing risky assumptions

Business Idea	Possible Risky Assumption	How to Test It (MVP)
Online tuition finder	Students will pay for online tutoring.	Ask 20 students and offer a trial class to see if they sign up.
Canteen pre-order form	Students will order snacks before break.	Test a Google Form for one week in real time.
Ajrak online catalogue	Customers will place orders through WhatsApp.	Share catalogue with 30 customers and record actual orders.
College van update system	Parents and students will check van updates daily.	Create a trial WhatsApp broadcast group for one route.

Notice that every test measures **what people actually do** (sign up, order, check), not just what they say.

WORKED EXAMPLE

Worked example: finding the riskiest assumption. Idea: a home-tutor booking service for Grade IX–X students in Sukkur.

Step 1, list assumptions:

1. Parents want home tutors for their children.
2. Good tutors in Sukkur are willing to join.
3. Parents will **pay a booking fee** through our service instead of finding tutors through relatives.
4. Parents will like our logo.

Step 2, ask of each: "If this is wrong, does the whole business fail?" Logo (4): no, easily changed. Parents wanting tutors (1): very likely true; many already hire tutors. Tutors joining (2): important, but easy to check by phoning a few tutors. Paying a booking fee (3): if parents refuse to pay, the business earns nothing, so **this is the riskiest assumption**.

Step 3, design a small MVP test: post the offer in 3 parent WhatsApp groups for one week, offer a first booking at a small fee, and **count how many actually pay**.

REMEMBER IT

Ask: "What would kill this idea?" The answer is usually the riskiest assumption. Test it **first**, cheaply, with real behaviour.

COMMON MISTAKE

Mistake: Testing easy, safe assumptions (colours, names) first and leaving the dangerous one for later.

Fix: Test the assumption that could make the whole business fail first. Also, do not trust only "Yes, I would use it". Measure actual sign-ups or orders.

اردو میں سمجھیں

ہر کاروباری خیال کچھ مفروضوں (Assumptions) پر قائم ہوتا ہے، یعنی وہ باتیں جنہیں ہم سچ سمجھتے ہیں مگر ابھی آزمایا نہیں۔ سب سے خطرناک مفروضہ (Riskiest Assumption) وہ ہے جو غلط نکلے تو پورا کاروبار ناکام ہو جائے۔ اسے شروع میں ہی چھوٹے ایم وی پی سے آزمانا چاہیے تاکہ وقت اور پیسہ بچ سکے۔

CHECK YOURSELF

1. Define assumption.
2. What is the risky assumption for the Ajrak online catalogue, and how is it tested?
3. Why test the riskiest assumption early?

Answers: 1. Something we believe to be true but have not tested yet. 2. Customers will place orders through WhatsApp; share the catalogue with 30 customers and record actual orders. 3. Because if it is wrong the whole business fails, so testing it early saves time and money.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.

2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.

6.11 Developing an MVP

LEARN IT

To develop an MVP, the team must **select only the essential features**. Extra features should be avoided at the beginning. A common mistake is **trying to build everything at once**. This makes the product **difficult, costly, and slow**.

An **MVP plan** should clearly explain: the **problem, target users, main feature, basic design, estimated cost, marketing method, feedback method, and improvement plan**.

Table 6.8: MVP planning checklist

MVP Element	Question to Ask	Example Answer
Problem	What problem are we solving?	Students wait too long at canteen.
Target users	Who will use it first?	Grade XII students in one college.
Main feature	What is the ONE feature that solves it?	Pre-order snacks before break.
Basic design	What steps are needed?	Menu, name, class, order time.
Cost	What free/low-cost tools to use?	Google Form or printed slips.
Marketing	How will early users hear about it?	WhatsApp message to class group.
Feedback	How will users respond?	Short feedback form after 3 days.

The eighth element, the **improvement plan**, answers "What will we change after feedback?" (you will see it in the case study in 6.12).

WORKED EXAMPLE

Worked example: filling the checklist for a new idea. Idea: a college notice reminder for parents in Hyderabad (from Table 6.1).

MVP Element	Answer
Problem	Parents miss college notices (fee dates, PTMs, holidays).
Target users	Parents of first-year (Class XI) students of one college.
Main feature	Send each notice to parents' phones on the same day.
Basic design	Photo of notice + 2-line summary in Urdu and Sindhi + date.
Cost	Free: WhatsApp broadcast list from the college office phone.
Marketing	Students give a printed slip to parents with the number to save.
Feedback	After 2 weeks, a 3-question voice-note or phone survey of 20 parents.
Improvement plan	If parents say notices come too late, send a reminder 1 day before each date.

Notice: only **one** main feature, a **free** tool, and a clear way to collect feedback.

REMEMBER IT

Mnemonic for the 8 elements: "People Try My Bakery; Cheap Muffins Find Improvement": Problem, Target users, Main feature, Basic design, Cost, Marketing, Feedback, Improvement plan.

COMMON MISTAKE

Mistake: Writing a list of ten "main features".

Fix: The checklist asks for the **ONE** feature that solves the problem. Everything else waits until the MVP proves people want it.

ایم وی پی بناتے وقت صرف ضروری خصوصیات چنیں؛ سب کچھ ایک ساتھ بنانے سے کام مشکل، مہنگا اور سست ہو جاتا ہے۔ ایم وی پی منصوبے میں مسئلہ، پہلے صارفین، ایک بنیادی خصوصیت، سادہ ڈیزائن، کم خرچ ٹولز، تشہیر کا طریقہ، رائے لینے کا طریقہ اور بہتری کا منصوبہ واضح ہونا چاہیے۔

CHECK YOURSELF

1. Why is building everything at once a mistake?
2. List the eight elements of an MVP plan.
3. In Table 6.8, what is the marketing method?

Answers: 1. It makes the product difficult, costly and slow. 2. Problem, target users, main feature, basic design, estimated cost, marketing method, feedback method, improvement plan. 3. A WhatsApp message to the class group.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Developing an MVP means:

- a. Choosing only essential features and a way to deliver them
- b. Hiring 40 engineers first
- c. Designing 12 themes
- d. Writing the TOS of a unicorn

MCQ 2. A feature is essential if:

- a. The core job fails without it
- b. A competitor has it
- c. It is AI
- d. It is animated

MCQ 3. Manual-behind-the-scenes is allowed if:

- a. The user still gets the job done (concierge MVP)
- b. You lie that it is fully automatic forever
- c. You skip users
- d. You skip ethics

MCQ 4. Time-box the build to:

- a. Days/weeks, not years
- b. A decade
- c. Until perfect
- d. Until the logo shines

MCQ 5. You still need:

- a. A way to measure the learning
- b. No metric
- c. Only likes
- d. Only press

MCQ 6. Extra features before the core works are:

- a. Waste
- b. Professionalism
- c. Required by BIEK
- d. HCI law

Extra short questions

1. Steps to develop an MVP from the notes.
2. Concierge MVP meaning.
3. Tools a student can use.
4. One ethics line.

Extra long question

1. Turn the notes-exchange idea into an MVP plan: features in/out, tools, 7-day schedule, metric, kill-criteria.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Core job, essential features, drop extras, pick tools, build, measure.
2. You perform the service by hand (WhatsApp) while pretending the “product” exists — to test demand.
3. Forms, sheets, WhatsApp, JazzCash, a landing page.
4. If you take money, deliver or refund; don't ghost testers.

Long

1. In: list+request. Out: chat/AI. Tools: sheet+group. Metric: 10 completed exchanges. Kill if <2. Schedule table.

6.12 Case Study: MVP for a Canteen Pre-Order System

LEARN IT

Problem: In a college, students wait in long lines at the canteen. Sometimes snacks finish before students reach the counter. A group wants to test a canteen pre-order idea.

Goal: Their aim is **NOT** to create a complete app immediately. Their aim is to test **whether students will place orders before break** and **whether the canteen can manage these orders**.

The approach

1. The team creates a **paper prototype** showing three steps: **select item, enter name, collect order**.
2. **Feedback** from the prototype suggests the **menu should be short** and **payment should be at collection time**.
3. For the actual MVP, the team uses a **Google Form**. Students select **one of five items**.
4. The canteen prepares these pre-orders. This MVP is tested for **three days only** to learn quickly.

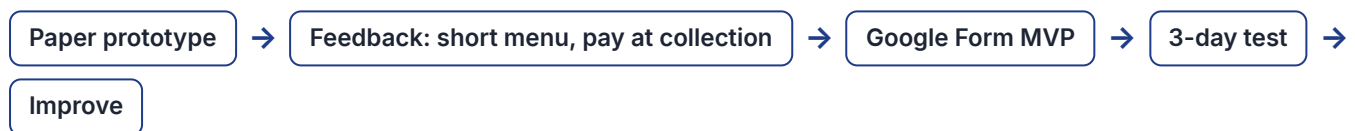


Table 6.9: MVP plan for canteen pre-order system

MVP Element	Decision
Features	Menu list, student name, class, item, and quantity.
Cost & Tools	Free Google Form or printed order slips.
Marketing	Class announcement and WhatsApp group message.
Testing period	Three days.
Feedback method	Ask students if they saved time and would use it again.
Improvement	Keep only 5 items; close orders 20 minutes before break.

Result: Students liked saving time, but **canteen staff found too many items difficult**. The team improved the MVP by **keeping only five items** and creating a **separate collection line**. This improvement is based on real feedback.

WORKED EXAMPLE

Worked example: trace the case study through the chapter's ideas.

Chapter idea	Where it appears in the case study
Real problem (6.3)	Long queues; snacks finish before students reach the counter.
Low-fidelity prototype (6.5)	Paper screens: select item, enter name, collect order.
Iteration (6.6)	Feedback led to a short menu and payment at collection.
MVP (6.8)	Google Form with one of five items, used for real orders.
Riskiest assumption (6.10)	Students will order before break, and the canteen can manage the orders.
Beachhead market (6.14)	One college canteen during break.

Why "close orders 20 minutes before break"? Staff need time to prepare. If orders arrive at the last minute, the pre-order line becomes as slow as the normal line, and the main benefit disappears.

REMEMBER IT

Remember the case study as **"5 items, 3 days, 20 minutes"**: five menu items, a three-day test, orders closing twenty minutes before break.

COMMON MISTAKE

Mistake: Thinking only the students' opinion matters.

Fix: The case study tested **two** things: whether students will order **and** whether the canteen can manage the orders. The staff's feedback (too many items) caused the key improvement.

اردو میں سمجھیں

ایک کالج میں طلبہ کینٹین کی لمبی قطاروں میں انتظار کرتے تھے۔ ٹیم نے فوراً ایپ بنانے کے بجائے پہلے کاغذی پروٹوٹائپ بنایا، پھر تین دن کے لیے گوگل فارم کو ایم وی پی کے طور پر آزمایا جس میں پانچ اشیاء تھیں۔ طلبہ کو وقت کی بچت پسند آئی مگر عملے کو زیادہ اشیاء مشکل لگیں، اس لیے صرف پانچ اشیاء رکھی گئیں، الگ قطار بنائی گئی اور بریک سے بیس منٹ پہلے آرڈر بند کیے گئے۔

CHECK YOURSELF

1. Which tool was used for the canteen MVP, and for how long was it tested?
2. What two things did the team want to test?
3. What two improvements were made after the result?

Answers: 1. A Google Form, tested for three days. 2. Whether students will place orders before break, and whether the canteen can manage these orders. 3. Keeping only five items and creating a separate collection line (orders also close 20 minutes before break).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The canteen case is there to:

- a. Show a full MVP story on a local problem
- b. Teach NAND
- c. Teach OSI
- d. Teach ER only

MCQ 2. The problem is:

- a. Queues / food running out
- b. GPUs
- c. Primary keys
- d. Figma auto-layout

MCQ 3. A sensible MVP is:

- a. Pre-order a few items for pickup, not a full restaurant OS
- b. Drone delivery in week 1
- c. AI chefs
- d. Blockchain loyalty

MCQ 4. The kitchen is a:

- a. User/constraint you must test
- b. A pie
- c. A stack
- d. An OSI hub

MCQ 5. If nobody pre-orders, the idea:

- a. Failed the risky assumption — good you learned cheap
- b. Needs more gradients
- c. Needs a CEO
- d. Needs C++

MCQ 6. Write the case as:

- a. Problem, users, assumption, MVP, test, result, next
- b. Only the logo
- c. Only careers
- d. Only Python lists

Extra short questions

1. Retell the case in 5 bullets.
2. Two user groups with different needs.
3. A metric.
4. A next step if it works.

Extra long question

1. Write the canteen case as a 10-mark structured answer with a tiny flow diagram (order → kitchen → pickup).

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Queue; students+staff+cooks; assume pre-order helps; form/WhatsApp MVP; measure wait/orders; iterate.
2. Students (speed); kitchen (simple tickets, not 40 modifiers).
3. Orders completed; average wait; food wasted.
4. Add payment; then maybe an app.

Long

1. Headings as above. Diagram three boxes. One risk: kitchen overwhelmed at 10:55. Mitigation: cap orders. Ethics: allergies.

6.13 Testing an MVP

LEARN IT

MVP testing means giving the basic working version to **real users** and **studying their response**. The team should collect **both numbers and comments**.

- **Numbers** (quantitative data) show how many users tried the MVP and how many may use it again.
- **Comments** (qualitative data) explain what users liked, disliked, or found confusing.

Testing should be **planned before the MVP is launched**. Decide:

1. **How many users** will test it.
2. **How long** the test will continue.
3. **What information** will be collected.

Table 6.10: Feedback during MVP testing

Type of Feedback	Question	Example Result
Number-based	How many users tried it?	40 students used the form.
Number-based	How many would use it again?	30 out of 40 said yes.
Comment-based	What was confusing?	Did not know order closing time.
Observation	Where did users face difficulty?	Forgot to enter class name.

Observation means the team watches users and notes difficulties themselves, even ones users do not mention.

WORKED EXAMPLE

Worked example: studying the results in Table 6.10.

1. **Repeat-use rate** = (users who would use it again ÷ users who tried it) × 100 = (30 ÷ 40) × 100 = **75%**.
2. **Did not want to use again** = 40 - 30 = **10 students** = (10 ÷ 40) × 100 = **25%**.
3. **Comment:** "Did not know order closing time" → **Fix:** write "Orders close 20 minutes before break" at the top of the form and in the WhatsApp message.
4. **Observation:** "Forgot to enter class name" → **Fix:** make the Class field **required** in the Google Form so it cannot be submitted empty.
5. **Decide what to fix first:** the closing-time problem stops people from ordering at all, so fix it first; the class-name problem only makes collection slower.

Three out of four users would come back, which is a promising sign; fixing the two problems could raise that number further.

REMEMBER IT

Data skill link. Feedback is useful only when it is studied: **Count** the responses → **Look** for repeated problems → **Decide** which improvement comes first. Remember "**C-L-D**: Count, Look, Decide."

COMMON MISTAKE

Mistake: Launching the MVP first and deciding later what to measure.

Fix: Plan the number of users, the test length and the questions **before** launch. Otherwise you may forget to record important data (for example, how many people opened the form but did not submit).

اردو میں سمجھیں

ایم وی پی کی جانچ کا مطلب ہے کہ بنیادی کام کرنے والا ورژن اصل صارفین کو دیا جائے اور ان کے ردعمل کا مطالعہ کیا جائے۔ اعداد بتاتے ہیں کتنے لوگوں نے استعمال کیا اور کتنے دوبارہ کریں گے، جبکہ تبصرے بتاتے ہیں کیا پسند آیا اور کیا الجھن تھی۔ جانچ کی منصوبہ بندی پہلے سے کریں، جوابات گنیں، بار بار آنے والے مسائل تلاش کریں اور پہلے سب سے اہم بہتری کریں۔

CHECK YOURSELF

1. Name the three types of feedback in Table 6.10.
2. If 50 people tried an MVP and 35 would use it again, what is the repeat-use rate?
3. Name the three things to decide before launching an MVP test.

Answers: 1. Number-based, comment-based, observation. 2. $(35 \div 50) \times 100 = 70\%$. 3. How many users will test it, how long the test will continue, and what information will be collected.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Testing an MVP means:

- a. Real users, real core task, collect numbers and comments
- b. Only your friends saying "cool logo"
- c. Only the teacher's mark
- d. Only a unit test of Python

MCQ 2. Vanity metrics are:

- a. Likes with no usage
- b. Completed jobs
- c. Repeat orders
- d. Time to complete

MCQ 3. You want both:

- a. Numbers and quotes
- b. Only numbers
- c. Only vibes
- d. Only press

MCQ 4. A failed test is useful when:

- a. You write what you will change or stop
- b. You hide it
- c. You add features at random
- d. You buy ads

MCQ 5. Sample of 5–20 users is:

- a. OK for class MVPs if honest
- b. A national census
- c. Useless always
- d. A DoS

MCQ 6. Don't help the user in the test if you are measuring:

- a. Whether the offer is understandable/usable
- b. Kindness in general life
- c. SQL skill
- d. NAND skill

Extra short questions

1. What to measure (book + good practice).
2. How to collect.
3. Bias: testing only your friends.
4. Then what?

Extra long question

1. Write a test plan for the canteen MVP: n, tasks, 3 metrics, 3 questions, pass/fail thresholds, ethics (food safety).

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Did they finish? Time? Would they repeat? Pay? Quotes of confusion.
2. Tally sheet + 3 interview questions after.
3. They are too nice — include one stranger in the line.
4. Iterate the MVP (or the assumption).

Long

1. n=15 lunch buyers. Task: order and collect. Metrics: % complete, wait, errors.
Questions: would you do this tomorrow? Thresholds: 70% complete. Don't poison anyone; allergens.

6.14 Beachhead Market ★ Golden

LEARN IT

A **beachhead market** is a **small and focused group of first users**. These users have a **clear need** and are **likely to try the product early**.

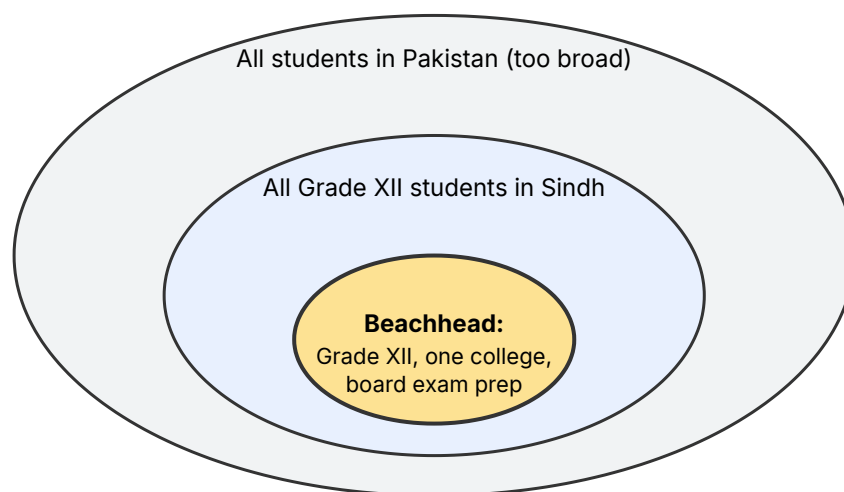
Instead of trying to serve everyone, entrepreneurs **first focus on one small group**. This helps them **test the MVP properly and improve it before expanding**.

The word comes from the army: a *beachhead* is the small piece of beach soldiers capture first when they land; from there they move inland. In business, your beachhead is the first small group you "win".

Characteristics of a good beachhead market

- It should be **specific and narrowly defined**.
- It should be **reachable** by the startup team.
- The users should have a **strong, immediate need** for the solution.

For example, targeting "**all students in Pakistan**" is too broad for a new study app. "**Grade XII students preparing for board exams in one college**" is much more focused and easier to reach.



Start with the small inner circle (beachhead), learn and improve, then expand outward.

Table 6.11: Choosing a focused beachhead market

Business Idea	Weak First Market	Better Beachhead Market
Study app	All students in Pakistan.	Grade XII students preparing for exams in one college.
Canteen pre-order	All canteens in the city.	One college canteen during lunch break.

Ajrak catalogue	All online shoppers.	Existing customers of one local shop.
Van update system	All transport users.	Students and parents of ONE specific college van route.

Choosing a small first market **does not mean the idea will remain small forever**. After learning from this group, the idea can be improved and expanded, even globally.

WORKED EXAMPLE

Worked example: test a proposed beachhead against the three characteristics. Idea: an app that reminds farmers about irrigation water turns (*warabandi*).

Proposed market	Specific?	Reachable?	Strong, immediate need?	Verdict
All farmers in Sindh	No	No (millions of people, many districts)	Varies	Weak
Farmers on one watercourse near Badin whose sons/daughters study in the team's college	Yes	Yes (through classmates)	Yes (missing a turn damages crops)	Good beachhead

Expansion path: one watercourse → nearby watercourses → the whole district → other districts.

REMEMBER IT

Remember S-R-U: Specific, Reachable, Urgent need. If you cannot meet your first users within a week, your market is too big.

COMMON MISTAKE

Mistake: Thinking a bigger first market means more success ("everyone can use my app").

Fix: A broad market is hard to reach and hard to learn from. Win a small group first, learn, improve, then expand.

بیج ہیڈ مارکیٹ (Beachhead Market) پہلے صارفین کا ایک چھوٹا اور مخصوص گروہ ہے جسے حل کی واضح اور فوری ضرورت ہو اور جو پراڈکٹ جلد آزمانے پر تیار ہو۔ سب کو ایک ساتھ خدمت دینے کے بجائے پہلے ایک چھوٹے گروہ پر توجہ دیں، جیسے "پاکستان کے تمام طلبہ" کے بجائے "ایک کالج کے بارہویں جماعت کے طلبہ"۔ وہاں سے سیکھ کر خیال کو بعد میں پھیلا یا جا سکتا ہے۔

EXAM TIP

Asked as "Define beachhead market and give its characteristics" or "Suggest a beachhead market for ...". Write the definition (small, focused group of first users with a clear need, likely to try early), the three characteristics (specific, reachable, strong immediate need), one weak-versus-better pair from Table 6.11, and the line that a small first market does not mean the idea stays small.

CHECK YOURSELF

1. Give the three characteristics of a good beachhead market.
2. What is the better beachhead market for the van update system?
3. Does choosing a small first market mean the business will stay small?

Answers: 1. Specific and narrowly defined; reachable by the team; users have a strong, immediate need. 2. Students and parents of ONE specific college van route. 3. No. After learning from the first group, the idea can be improved and expanded.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A beachhead market is:

- a. A small focused first group of users
- b. The whole country on day 1
- c. Random Twitter
- d. Every age/city/income at once

MCQ 2. You pick them because:

- a. They feel the pain strongly and you can reach them
- b. They are richest always
- c. They are abroad always
- d. They hate the product

MCQ 3. This is ★: define, why small, one example.

- a. The recipe
- b. Only the D-Day story without CS
- c. Only Python
- d. Only ER

MCQ 4. XI CS students in one college as first users of a notes app is a:

- a. Beachhead
- b. Global TAM
- c. OSI layer
- d. NAND net

MCQ 5. After the beachhead works you:

- a. Expand to the next similar group
- b. Immediately advertise on TV nationwide
- c. Quit
- d. Rewrite in assembly

MCQ 6. A beachhead that is too broad is:

- a. Hard to test and to talk to
- b. Ideal
- c. Required
- d. A median

Extra short questions

1. Define beachhead market.
2. Why not "all Pakistan" first?
3. Example from college.
4. How it links to MVP.

Extra long question

1. For a canteen pre-order, choose a beachhead (who, where, when), why they are best, and who you refuse at first. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Small, focused first users with a clear need, likely to try.
2. You cannot learn; message becomes vague; cost explodes.
3. Hostel Block B, not all Karachi youth.
4. The MVP is aimed at that beachhead, not every imaginary user.

Long

1. XI 10:00 break, one stall. They queue worst. Refuse staff catering and evening shift until the kitchen process works. Success: 20 repeating students.

6.15 & 6.16 Complete Student Project & Ethical Use

LEARN IT

You will work in **groups** and complete one small entrepreneurship project. The project should be **simple, local, and possible in the classroom**.

The purpose is to practise the **complete journey**: problem → prototype → feedback → MVP plan → risky assumption → beachhead market.

Suggested project ideas: homework and test reminder system, college notice organizer, canteen pre-order system, small shop WhatsApp catalogue, college van update system, lost and found item reporting system, study notes organizer.

Table 6.12: Student project sheet (template)

Item	Your group's answer
Problem statement	
Target users	
Business idea	
Low-fidelity prototype description	
Feedback from users	
Improvement made after feedback	
MVP plan	
Riskiest assumption	
How to test the riskiest assumption	
Beachhead market	
Marketing method for first users	

WORKED EXAMPLE

Worked example: a completed project sheet (Lost and Found system).

Item	Answer
Problem statement	Students of our college in Mirpurkhas lose items (bottles, calculators, ID cards) and rarely get them back because there is no central place to report them.
Target users	Students and staff of our college.
Business idea	A Lost and Found WhatsApp group plus a Google Form to report lost or found items.
Low-fidelity prototype	Two paper screens: "Report lost item" (item, colour, place, phone of class rep) and "Found items list" with sketches.
Feedback from users	6 of 8 testers asked for photos; 3 did not want their own phone number shown.
Improvement made	Added a photo upload; replaced personal numbers with "collect from the library desk".
MVP plan	Google Form + one WhatsApp group run by two volunteers; free; tested for two weeks.
Riskiest assumption	Students who find items will take the trouble to report them.
How to test it	Count how many "found" reports arrive in two weeks compared with "lost" reports.
Beachhead market	Grade XII students of our college.
Marketing method	Announcement in morning assembly, a poster at the library and a message in each class group.

Table 6.13: Rubric for project assessment

A **rubric** is a marking guide that describes what excellent, satisfactory and weak work looks like for each part of a project.

Criteria	Excellent	Satisfactory	Needs Improvement
Problem ID	Clear, local, and important.	Understandable but needs detail.	Vague or too broad.

Prototype	Shows clear flow and features.	Shows the basic idea.	Incomplete or unclear.
Feedback & Fixes	Collected properly; changes clearly based on feedback.	Collected but analysis is weak; some changes made.	No clear feedback collected; changes not explained.
MVP Plan	Essential features and clear testing plan.	Basic features but needs testing clarity.	Confused with the final product.

REMEMBER IT

Use the rubric **before** you submit: read your project sheet row by row and ask, "Is this Excellent yet?" For "Feedback & Fixes", always write **which** feedback caused **which** change.

COMMON MISTAKE

Mistake: Writing an MVP plan that lists every feature of the dream app. The rubric marks this as "Confused with the final product".

Fix: Keep the MVP to the essential feature(s) and add a clear testing plan (how many users, how long, what you will measure).

اردو میں سمجھیں

طلبہ گروپوں میں ایک چھوٹا، مقامی اور آسان منصوبہ مکمل کرتے ہیں جس میں مسئلے سے پروٹوٹائپ، رائے، ایم وی پی منصوبہ، سب سے خطرناک مفروضہ اور بیچ ہیڈ مارکیٹ تک پورا سفر شامل ہوتا ہے۔ منصوبے کو جانچنے کے لیے روبرک (Rubric) استعمال ہوتا ہے جو مسئلے کی پہچان، پروٹوٹائپ، رائے و بہتری اور ایم وی پی منصوبے کو بہترین، تسلی بخش یا بہتری کی ضرورت کے درجوں میں پرکھتا ہے۔

CHECK YOURSELF

1. Name four suggested project ideas.
2. What does "Excellent" look like for Problem ID in the rubric?
3. Which rubric level says "Confused with the final product"?

Answers: 1. Any four: homework and test reminder system, college notice organizer, canteen pre-order system, small shop WhatsApp catalogue, college van update system, lost and found reporting system, study notes organizer. 2. Clear, local, and important. 3. Needs Improvement (for the MVP Plan).

LEARN IT

Digital entrepreneurship should be **responsible**. When collecting user feedback, you must respect:

- **Privacy**: people's personal information belongs to them.
- **Truthfulness**: report results honestly; do not make up numbers or hide bad feedback.
- **Fairness**: treat all users equally.

You should **not collect unnecessary personal information** or **share phone numbers or photos without permission**.

Equity point

A good digital solution should consider users who have **weak internet**, **limited devices**, or **language difficulty**. A design with **large buttons**, **simple Urdu/Sindhi instructions**, and **low data usage** helps ensure **equal access**. *Equity* means making sure everyone, including people with fewer resources, can use the solution.

WORKED EXAMPLE

Worked example: fix an unethical feedback form. A team's canteen feedback form asks:

1. Full name, father's name, CNIC/B-form number, home address. → **Problem**: unnecessary personal data. **Fix**: ask only class (or nothing personal at all).
2. "Upload your selfie." → **Problem**: photos are not needed and could be misused. **Fix**: remove.
3. The team plans to post all respondents' phone numbers in the class group. → **Problem**: sharing numbers without permission. **Fix**: never share; keep responses private to the team.
4. The form is in English only and has a 5 MB video at the top. → **Problem**: excludes users with language difficulty or weak internet. **Fix**: add Urdu/Sindhi instructions and remove the heavy video.

When presenting, the team reports "30 out of 40 would use it again" exactly, without rounding it up to "almost everyone". That is truthfulness.

REMEMBER IT

Ask yourself "P-T-F-E": Is it **Private**, **Truthful**, **Fair**, and **Equal-access**?

COMMON MISTAKE

Mistake: "More data is always better, so let's collect everything."

Fix: Collect only what you need to answer your test questions. Extra personal data creates risk for users and gives you no benefit.

ڈیجیٹل کاروبار ذمہ داری سے کرنا چاہیے۔ رائے جمع کرتے وقت لوگوں کی رازداری (Privacy)، سچائی اور انصاف کا خیال رکھیں؛ غیر ضروری ذاتی معلومات نہ لیں اور اجازت کے بغیر فون نمبر یا تصاویر شیئر نہ کریں۔ اچھا حل کمزور انٹرنیٹ، محدود موبائل اور زبان کی مشکل والے لوگوں کا بھی خیال رکھتا ہے، جیسے بڑے بٹن، آسان اردو یا سندھی ہدایات اور کم ڈیٹا کا استعمال۔

CHECK YOURSELF

1. Name the three values students must respect when collecting feedback.
2. Give two things students must not do with users' information.
3. List three design features that support equal access.

Answers: 1. Privacy, truthfulness, fairness. 2. Collect unnecessary personal information; share phone numbers or photos without permission. 3. Large buttons, simple Urdu/Sindhi instructions, low data usage.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The student project should be:

- a. Small, local, possible in class time
- b. A full bank app
- c. A new OS
- d. A satellite

MCQ 2. It should include:

- a. Problem, idea, prototype/MVP, test, ethics
- b. Only a logo
- c. Only careers
- d. Only a Python hello

MCQ 3. Ethical use of digital tools means:

- a. Consent, no harassment, no leaked data, honest claims
- b. Scrape classmates' photos
- c. Fake testimonials
- d. Spam the college group

MCQ 4. The rubric rewards:

- a. A real test, not a fantasy pitch deck
- b. Unicorn valuation
- c. English poetry only
- d. NAND counts

MCQ 5. Safe tools:

- a. Forms/docs you have a right to use; no piracy
- b. Cracked Adobe
- c. Stolen question papers
- d. Phishing kits

MCQ 6. 6.15 and 6.16 together remind you:

- a. Build something small and don't harm people
- b. Skip users
- c. Skip tests
- d. Skip the write-up

Extra short questions

1. List the project pieces the book wants.
2. Two ethical rules for digital tools.
3. What "possible" means.
4. A fail on ethics even if the idea is clever.

Extra long question

1. Write a one-page project proposal that would score well: 8 headings including ethics and a 7-day test. Invent a local idea that is not the canteen copy-paste.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Team, local problem, idea, prototype, MVP or test, beachhead, results, ethics, reflection.
2. Permission for photos; no copyright theft; no unsafe personal data in public sheets.
3. You can test this week with people you can walk to.
4. Secretly recording students in the canteen.

Long

1. e.g. lab-equipment booking. Headings match rubric. Ethics: no shame list of late students on a public wall. Metric: 10 bookings honoured.

Entrepreneurship in the Digital Age

16 single-page lessons · 5 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- **Entrepreneur:** a person who identifies a problem, thinks of a solution, and turns it into a product or service; observes people, takes calculated risks, improves step by step.
- **Entrepreneurship:** the process of planning, starting, testing and improving a business idea until it becomes useful (Problem → Solution → Product).
- **Digital age:** entrepreneurs use accessible digital tools (social media, forms, websites, WhatsApp, digital payments) to start and test ideas quickly with low investment; often no coding is needed.
- A digital idea must solve a **real problem for real users**, not just sound modern (Table 6.1 examples: notices, embroidery shop, test dates, canteen queues).
- **Business ideas** come from observing daily life and talking directly to users: Observe → Ask Users → Select One Problem → Suggest Solution → Test Early. Start with a small, clear problem.
- **Prototype:** an early model used to show and test the design of an idea; not the final product; makes the idea visible, finds mistakes early, saves time and money.
- **Fidelity** is how close a prototype is to the final product: Low (paper), Mid (PowerPoint/Canva/Figma, tests navigation), High (realistic, Figma/HTML, only after the idea is tested).
- **Development cycle:** Design → Build → Test → Iterate, repeated (iteration). Feedback is crucial; a prototype fails only if useful feedback is ignored.
- **MVP:** the simplest working version with only essential features (Minimum) that still solves the main problem (Viable); it tests real use and market demand.
- **Prototype vs MVP:** prototype asks "Does it look right?"; MVP asks "Will people actually use it?" A prototype may not work; an MVP must.
- **Riskiest assumption:** the untested belief that, if wrong, will make the business fail; test it early with real behaviour.
- **MVP plan:** problem, target users, ONE main feature, basic design, cost, marketing, feedback, improvement plan. Do not build everything at once.
- **Canteen case study:** paper prototype → Google Form MVP with 5 items for 3 days → staff found too many items hard → kept 5 items, separate line, orders close 20 minutes before break.
- **MVP testing:** plan users, duration and data in advance; collect numbers (40 tried, 30 would return = 75%) and comments/observations; count, look for repeated problems, fix the most important first.
- **Beachhead market:** a small, focused, reachable group of first users with a strong, immediate need; win it first, then expand.
- **Ethics and equity:** respect privacy, truthfulness and fairness; collect no unnecessary data; share nothing without permission; design for weak internet, limited devices and Urdu/Sindhi speakers.

Key terms

Term	Meaning	اردو
Entrepreneur	A person who identifies a problem, thinks of a solution, and turns it into a product or service.	کاروباری شخص
Entrepreneurship	The process of planning, starting, testing and improving a business idea.	کاروبار کاری

Calculated risk	A risk taken after carefully thinking about the possible loss and keeping it small.	سوچا سمجھا خطرہ
Digital entrepreneurship	Using digital tools to start, test, improve and grow a business idea.	ڈیجیٹل کاروبار کاری
Business idea	A proposed product or service that solves a real problem for real users.	کاروباری خیال
Target users	The people who will use the product first.	ہدف صارفین
Prototype	An early model of a product, service, website or app made to test the idea and collect feedback.	ابتدائی نمونہ
Fidelity	How close a prototype is to the final product.	اصل سے مطابقت کا درجہ
Low-fidelity prototype	A quick, rough paper model showing the main idea and flow.	کم تفصیل والا نمونہ
Mid-fidelity prototype	A clearer model in PowerPoint, Canva or Figma showing layout and navigation without final graphics.	درمیانی تفصیل والا نمونہ
High-fidelity prototype	A detailed model close to the final product with realistic colours, buttons, data and working features.	زیادہ تفصیل والا نمونہ
Wireframe	A simple outline drawing of a screen showing where items go, without colours or pictures.	اسکرین کا خاکہ

CHAPTER 6 EXAM • CLOSED BOOK

Entrepreneurship in the Digital Age

75 marks • 1 hour 15 minutes • no notes • no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 16 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. 15 + 30 + 30 = 75. A 5-mark / 10-mark answer needs Define • Explain • Example • Diagram • Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. An entrepreneur is best described as a person who:
 - a. only invests money in big companies
 - b. identifies a problem and tries to turn a solution into a product or service
 - c. always works from a large office
 - d. only writes computer programs
2. In prototyping, "fidelity" means:
 - a. the cost of the prototype
 - b. the number of users
 - c. how close the prototype is to the final product
 - d. the speed of the internet
3. Which of these is a low-fidelity prototype?
 - a. Hand-drawn app screens on paper
 - b. A working demo website of an order form
 - c. Clickable screens in Figma
 - d. An app published on the Play Store
4. A mid-fidelity prototype is mainly useful for testing:
 - a. online payments
 - b. screen-to-screen navigation
 - c. market demand over three days
 - d. investor interest
5. The correct order of the prototype development cycle is:
 - a. Build, Design, Iterate, Test
 - b. Test, Build, Design, Iterate
 - c. Design, Test, Build, Iterate
 - d. Design, Build, Test, Iterate
6. Creating, testing, improving and testing a prototype again is called:
 - a. iteration
 - b. incubation
 - c. marketing
 - d. assumption
7. In MVP, the letter "V" stands for:
 - a. Visual
 - b. Valuable
 - c. Viable
 - d. Virtual
8. Which is the best MVP for a canteen pre-order idea?
 - a. A full app with login, payment, delivery tracking and discounts
 - b. A Google Form used for real orders for three days
 - c. Paper sketches shown to five friends
 - d. A colourful poster of the menu
9. A prototype mainly answers which question?
 - a. Does it look right and make sense?
 - b. Will people pay for it every month?
 - c. How big is the national market?
 - d. How much profit will we make?

- .0. The riskiest assumption is:
- a. the cheapest feature to build
 - b. the colour scheme of the app
 - c. the name of the company
 - d. the belief that, if wrong, can make the whole business fail
- .1. Which is the best beachhead market for a new study app?
- a. All students in Pakistan
 - b. All online shoppers
 - c. Grade XII students preparing for board exams in one college
 - d. All transport users
- .2. Which is NOT a characteristic of a good beachhead market?
- a. Specific and narrowly defined
 - b. Reachable by the startup team
 - c. Users have a strong, immediate need
 - d. As large as possible from the first day
- .3. "30 out of 40 said they would use it again" is which type of feedback?
- a. Number-based
 - b. Comment-based
 - c. Observation
 - d. Rubric-based
- .4. Which is an ethical practice when collecting user feedback?
- a. Collect CNIC numbers of all users
 - b. Share users' phone numbers in the class group
 - c. Collect only necessary information and share nothing without permission
 - d. Change results so the idea looks better
- .5. To support users with weak internet and language difficulty, a design should use:
- a. heavy videos on every screen
 - b. English-only instructions
 - c. features that need the latest phone
 - d. large buttons, simple Urdu/Sindhi instructions and low data usage

Section B — Short questions (10 × 3 = 30)

1. Define entrepreneur and entrepreneurship. How are they different?
2. What is meant by entrepreneurship in the digital age? Give two local examples that need no coding.
3. Write the five questions you should ask to find a problem in daily life.
4. Why should a student choose a small problem at the start? Give one example of a too-large and one of a suitable problem.
5. Define prototype and give three reasons why it is useful.
6. Differentiate between low-, mid- and high-fidelity prototypes, with one example of each.
7. Define MVP and explain the words "Minimum" and "Viable".

8. What is the riskiest assumption? Give the risky assumption and its test for the college van update system.
9. An MVP was tried by 40 students and 30 said they would use it again. Calculate the repeat-use rate and name two other kinds of feedback to collect.
10. State the three characteristics of a good beachhead market and give one weak and one better first market for an Ajrak catalogue.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Explain the prototype development cycle with a labelled diagram. Use the ClassBuddy class activity to show what happens at each stage, and explain why negative feedback does not mean the prototype has failed.
2. Compare a prototype and an MVP under five headings (purpose, working status, users, cost, example). Use the canteen pre-order system as your example and state the key question each one answers.
3. **Plan an MVP for a college van update system** for a college in Hyderabad, where students and parents often do not know when the van will arrive. Fill in all eight elements of the MVP planning checklist, and also state the riskiest assumption, how to test it, and the beachhead market.
4. **Plan an MVP for a small Sindhi topi and Ajrak shop in Hala** that wants online customers. Describe a low-fidelity prototype, the MVP, its riskiest assumption and test, the beachhead market, the marketing method, and two ethical or equity points the shop must follow.
5. A team tested a Lost and Found MVP (Google Form + WhatsApp group) in a college of 60 Grade XII students for two weeks. 45 students tried it; 36 of them said they would use it again; 12 of the users said they did not know where to collect found items; the team observed that many reports had no photo. (a) Calculate the percentage of the class who tried it, the repeat-use rate among users, and the percentage of users confused about collection. (b) Classify each piece of feedback by type. (c) Decide which improvement to make first and justify it.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **6.1 Entrepreneur and Entrepreneurship.** Answers: 1. A process (planning, starting, testing and improving a business idea). 2. Observing people and listening to their needs; taking calculated risks; improving the idea step by step. 3. A risk taken after thinking carefully about the possible loss and keeping that loss small.
2. **6.2 Entrepreneurship in the Digital Age.** Answers: 1. Any four of: mobile phones, social media, websites, online forms, digital payments, maps, videos, simple apps. 2. Because a small idea can be tested quickly and cheaply, so if users do not like it the team can change it before spending too much money. 3. Selling handmade cards through Instagram, or a WhatsApp catalogue for Ajrak.
3. **Table 6.1 Local examples of digital entrepreneurship.** Answers: 1. A simple catalogue with an order form on Facebook or WhatsApp. 2. Students and teachers. 3. Because it must solve a real problem for real users; a weak problem gives a weak business idea.
4. **6.3 From Problem to Business Idea.** Answers: 1. Observe, Ask Users, Select One Problem, Suggest Solution, Test Early. 2. It is too large and unclear; small, clear problems are easier to test and improve. 3. Notes are scattered in many places (WhatsApp, email, photocopies, class groups), not that students are careless.
5. **6.4 Prototype.** Answers: 1. No. It is an early model made to test the idea and collect feedback. 2. Any four: paper sketch, cardboard model, clickable PowerPoint, Canva design, Figma wireframe, simple website layout, sample service process chart. 3. Any two: a button is hard to find, a screen has too much text, a feature is missing.
6. **6.5 Types of Prototypes (Low, Mid, High-Fidelity).** Answers: 1. How close the prototype is to the final product. 2. Mid-fidelity (e.g. clickable screens in PowerPoint or Figma). 3. It is cheap, fast and easy to change, and it protects them from building a full app before knowing whether users need it.
7. **6.6 Prototype Development Cycle (Design, Build, Test, Iterate).** Answers: 1. The repeating process of creating, testing, improving and testing a prototype again. 2. The Test stage. 3. To make the idea visible (not to make it beautiful).
8. **6.7 Class Activity: Create a Prototype.** Answers: 1. 7 out of 10, which is 70%. 2. Today's classes, Add homework, Show reminder. 3. A 'Weekly View' button was added.
9. **6.8 Minimum Viable Product (MVP).** Answers: 1. Minimum Viable Product. 2. Usable enough to actually solve the main problem. 3. "Will people actually use this solution?"
10. **6.9 Difference between Prototype and MVP.** Answers: 1. The MVP. 2. Prototype: paper screens for a canteen order system. MVP: a Google Form used for real canteen orders for three days. 3. "Does it look right and make sense?"
11. **6.10 Identifying the Riskiest Assumption.** Answers: 1. Something we believe to be true but have not tested yet. 2. Customers will place orders through WhatsApp; share the catalogue with 30 customers and record actual orders. 3. Because if it is wrong the whole business fails, so testing it early saves time and money.
12. **6.11 Developing an MVP.** Answers: 1. It makes the product difficult, costly and slow. 2. Problem, target users, main feature, basic design, estimated cost, marketing method, feedback method, improvement plan. 3. A WhatsApp message to the class group.
13. **6.12 Case Study: MVP for a Canteen Pre-Order System.** Answers: 1. A Google Form, tested for three days. 2. Whether students will place orders before break, and whether the canteen can manage these orders. 3. Keeping only five items and creating a separate collection line (orders also close 20 minutes before break).

14. **6.13 Testing an MVP.** Answers: 1. Number-based, comment-based, observation. 2. $(35 \div 50) \times 100 = 70\%$. 3. How many users will test it, how long the test will continue, and what information will be collected.
15. **6.14 Beachhead Market.** Answers: 1. Specific and narrowly defined; reachable by the team; users have a strong, immediate need. 2. Students and parents of ONE specific college van route. 3. No. After learning from the first group, the idea can be improved and expanded.
16. **6.15 & 6.16 Complete Student Project & Ethical Use.** Answers: 1. Any four: homework and test reminder system, college notice organizer, canteen pre-order system, small shop WhatsApp catalogue, college van update system, lost and found reporting system, study notes organizer. 2. Clear, local, and important. 3. Needs Improvement (for the MVP Plan).

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. b | 7. c | 10. d | 13. a |
| 2. c | 5. d | 8. b | 11. c | 14. c |
| 3. a | 6. a | 9. a | 12. d | 15. d |

Short questions — model answers

1. An **entrepreneur** is a person who identifies a problem, thinks of a solution and tries to change that solution into a product or service. **Entrepreneurship** is the process of planning, starting, testing and improving a business idea, involving creativity, problem-solving, teamwork, communication and calculated risk. The entrepreneur is the person; entrepreneurship is the process.
2. It means using digital tools such as mobile phones, social media, websites, online forms, digital payments, maps, videos and simple apps to start, test, improve and grow a business idea. Examples: a student selling handmade cards through Instagram; a shopkeeper making a WhatsApp catalogue for Ajrak.
3. What problem do people face? Who faces this problem? How often does it happen? What do people currently do to solve it? Can technology help in a simple way?
4. Small, clear problems are easier to test and improve. Too large: "improving the whole education system". Suitable: "helping Grade XII students track practical file submission dates".
5. A prototype is an early model of a product, service, website or app that shows how the idea may look or work; it is not the final product and is made to test the idea and collect feedback. Useful because it makes the idea visible, helps find mistakes early (hard-to-find buttons, too much text, missing features), and saves time and money before final coding.
6. Low-fidelity: quick, rough paper model showing main idea and flow, e.g. hand-drawn homework-reminder screens. Mid-fidelity: clearer model in PowerPoint/Canva/Figma showing layout and navigation without final graphics, e.g. clickable PowerPoint screens. High-fidelity: detailed model close to the final product with realistic colours, buttons, sample data and working features, e.g. a working demo of a small order-form website.
7. An MVP (Minimum Viable Product) is the simplest working version of a product with only the most important features, used to test the idea with real users. Minimum: only the

essential features, no extras. Viable: usable enough to actually solve the main problem.

8. It is the belief that can make the entire business fail if it turns out to be wrong. For the van update system: "Parents and students will check van updates daily." Test: create a trial WhatsApp broadcast group for one route.
9. Repeat-use rate = $(30 \div 40) \times 100 = 75\%$. Other feedback: comment-based (e.g. "What was confusing?") and observation (e.g. where users faced difficulty, such as forgetting to enter class name).
10. Specific and narrowly defined; reachable by the team; users have a strong, immediate need. Ajrak catalogue: weak = all online shoppers; better = existing customers of one local shop.

Long questions — model answers

1. **Prototype development cycle.** Draw four boxes in a loop: Design → Build → Test → Iterate → back to Design. Prototype development is not one-time; repeating the cycle is called iteration.
 - **Design:** understand the problem and plan what the prototype shows. ClassBuddy: interview 10 students; 7 of 10 forget tasks, so show a daily task list; reminders are needed before leaving home, so add a morning notification; students want simplicity, so use large buttons and clear labels.
 - **Build:** create the first version to make the idea visible (not beautiful). ClassBuddy: three paper screens (Today's classes, Add homework, Show reminder) joined by arrows.
 - **Test:** show it to users and observe. ClassBuddy: five classmates try "add one homework task" while the observer watches silently.
 - **Iterate:** improve from feedback. ClassBuddy: larger, centred Add Homework button; subject name and due date fields; a Weekly View button.

Negative feedback is information that shows what to fix; the prototype fails only if the team ignores useful feedback.

2. **Prototype vs MVP.** Purpose: prototype shows how the idea may look or work; MVP tests whether users will actually use it. Working status: prototype may not work (sketch or model); MVP must work at least in a basic way. Users: prototype, a small group giving design feedback; MVP, early real users in daily life. Cost: prototype very low; MVP low but may need basic development. Example: prototype = paper screens for a canteen order system; MVP = Google Form used for real canteen orders for three days. Key questions: prototype, "Does it look right and make sense?"; MVP, "Will people actually use it and find value in it?"

3. **MVP plan: college van update system (Hyderabad).**

- Problem: students and parents do not know when the van will arrive, so students wait outside for a long time or miss the van.
- Target users: students and parents on one van route.
- Main feature: the driver or a route monitor sends a short "Van left / 5 minutes away" message.

- Basic design: one WhatsApp broadcast message with route stop name, time and status; voice note option for parents who cannot read.
- Cost: free, using a WhatsApp broadcast list on the monitor's phone.
- Marketing: a message in the class WhatsApp group and a printed slip for parents with the number to save.
- Feedback: after one week, a 3-question Google Form or phone call: Did you check the updates daily? Did you wait less? Would you continue?
- Improvement plan: if messages arrive too late, send them earlier; if many parents do not read, switch to voice notes; then expand to a second route.
- Riskiest assumption: parents and students will check van updates daily. Test: trial broadcast group for one route for one week, counting how many people read or reply to messages each day.
- Beachhead market: students and parents of ONE specific college van route.

4. MVP plan: topi and Ajrak shop in Hala.

- Low-fidelity prototype: paper catalogue pages, one per product, with a sketch, price, size/type and "Message to order"; shown to 5 customers to see if information is clear.
- MVP: a WhatsApp Business catalogue with real photos and prices for 10–15 best-selling items; orders by chat; payment by cash on delivery or JazzCash/Easypaisa.
- Riskiest assumption: customers will actually place orders through WhatsApp. Test: share the catalogue with 30 existing customers and record actual orders for two weeks.
- Beachhead market: existing customers of this one shop (people who have bought before and whose numbers the shop already has with permission).
- Marketing: broadcast message to existing customers, a QR code at the shop counter, posts in local Facebook groups.
- Ethics: share no customer numbers or photos without permission; give truthful prices, sizes and photos. Equity: simple Urdu/Sindhi descriptions and compressed photos for low data use.
- Improvement plan: drop designs nobody orders, add more of the popular ones, then expand to customers in Hyderabad and Karachi.

5. Lost and Found MVP analysis.

- (a) Tried: $(45 \div 60) \times 100 = 75\%$ of the class. Repeat-use rate: $(36 \div 45) \times 100 = 80\%$. Confused about collection: $(12 \div 45) \times 100 = 26.7\%$ (about 27%).
- (b) "45 tried" and "36 would use again": number-based. "Did not know where to collect": comment-based (12 users said it). "Many reports had no photo": observation.
- (c) Fix the collection confusion first: it is a repeated problem (more than a quarter of users) and it stops the system's main purpose, which is returning items to owners. Add "Collect found items from the library desk" at the top of the form and pin it in the WhatsApp group. Next, make the photo field strongly encouraged (or required for found items), since photos help owners recognise items. Re-test for another week and compare the numbers.

Self-assessment: I can...

- ☐ define entrepreneur and entrepreneurship and describe an entrepreneur's habits.
- ☐ use the five questions and the steps of Table 6.2 to turn a problem into an idea.
- ☐ compare low-, mid- and high-fidelity prototypes.
- ☐ turn user feedback into problems and improvements, as in Table 6.5.
- ☐ identify the riskiest assumption of an idea and design a test for it.
- ☐ choose a specific, reachable beachhead market.
- ☐ explain how digital tools let me start and test an idea cheaply, with local examples.
- ☐ define a prototype and list its benefits and possible forms.
- ☐ draw and explain the Design–Build–Test–Iterate cycle.
- ☐ define an MVP and explain how it differs from a prototype.
- ☐ fill in an MVP plan and analyse MVP test results with percentages.
- ☐ collect feedback ethically and design for equal access.